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Hierarchical model predictive control-guided reinforcement learning for energy management of hybrid-electric distributed propulsion aircraft

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ABSTRACT

This paper proposes a deployable hierarchical control architecture for mission-level energy management in a series hybrid-electric distributed propulsion (DEP) aircraft, explicitly balancing safety and long-horizon optimality. The architecture integrates model predictive control (MPC) with a reinforcement learning-based Soft Actor-Critic (SAC) agent in a phase-adaptive hierarchical structure. MPC ensures conservative, constraint-aware authority during safety-critical phases (takeoff and climb), while the SAC agent provides fuel- and energy-optimal power allocation in less constrained regimes, particularly cruise. Outputs are continuously blended using flight-phase-dependent weights rather than hard switching, with automatic fallback to MPC-only operation upon SAC degradation. The SAC policy, trained off-policy over full missions, generalizes across conditions while strictly enforcing battery state-of-charge and thermal constraints. The energy management is decoupled from trajectory regulation, avoiding online constrained or optimal inner-loop propulsion or speed control. Aircraft velocity evolves naturally from thrust–drag dynamics, guided by a simple fixed-structure feedback controller enforcing predefined phase-dependent speed targets, preserving physical consistency without online constrained optimization. Results demonstrate that the hierarchical MPC-guided SAC framework achieves superior performance across stability, tracking, and efficiency, yielding an overall score of 0.88. By maintaining small SAC corrections between 0.3 and 0.4 in magnitude, the system ensures robust transitions between control authorities. For a 0.38-hour reference mission, the controller leverages low-demand phases for in-flight battery recharging, delivering 9.79 kWh at a peak power of 103 kW to maximize SOC recovery. Compared to MPC-only benchmarks, the strategy achieves a 5.75% reduction in fuel consumption and a 5.7% decrease in CO_2 emissions for both Jet-A and sustainable aviation fuel (SAF). These findings establish that the integrated framework provides a robust, real-time solution for reducing emissions and extending battery longevity without compromising flight performance.

1. Introduction

Hybrid-electric propulsion (HEP) has emerged as a promising approach for improving the efficiency, sustainability, and operational flexibility of future aircraft by combining gas turbines with electrically driven propulsion and onboard energy storage. By decoupling power generation from thrust production, HEP architectures introduce new degrees of freedom in energy generation, distribution, and utilization across the mission. Among these architectures, distributed electric propulsion (DEP) offers particularly strong potential by enabling aerodynamic–propulsive integration benefits such as drag reduction, enhanced lift, and partial electrification of cruise, as demonstrated in recent experimental platforms [1,2]. However, these advantages introduce a fundamentally more complex system-level control problem than that of conventional propulsion systems. Rather than commanding thrust through a small number of centralized engines, HEP aircraft require continuous power allocation across heterogeneous energy sources while satisfying tightly coupled aerodynamic, propulsion, thermal, and energy storage constraints. Energy management decisions directly influence thrust generation, aerodynamic efficiency, fuel consumption, battery state-of-charge sustainability, and thermal loading of motors and power electronics, with strong dependence on flight regime. These interactions vary significantly between high-power takeoff and climb, energy-efficient cruise, and low-demand or regenerative descent, and must be coordinated over the full mission [3–5]. Consequently,

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Nomenclature

m	Mass [kg]
$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	Moments of inertia [kg·m^2]
g	Gravity [m/s^2]
ρ	Air density [kg/m^3]
S	Wing area [m^2]
b	Wingspan [m]
$\bar{c}$	Mean aerodynamic chord [m]
R, L, k_t, k_e, J	Motor damping parameters
k_T, k_Q, k_v	Propulsion coefficients
$y_{pos,j}, z_{pos,j}$	Motor positions [m]
C_T, η_{wash}	Prop-wash parameters
T	Total thrust [N]
P	Power required [W]
V	Freestream velocity[m/s]
n	Number of motors
A_{disk}	Propeller disk area per motor [m^2]
$C_{X0}, C_{X\alpha,\ldots}, C_{n\delta r}$	Aerodynamic coefficients
$u,\ v,\ w$	Body-fixed velocities[m/s]
p, q, r	Angular rates [rad/s]
V_j	Motor voltages [V]
a	Axial induction factor
P_{shaft}	Shaft power [W]
$\mathscr{H}$	Entropy term
SFC	Specific fuel consumption [kg/kWh]
P_{turb}	Turbine power output [W]
N	Rotational speed [RPM]
Q_{max}	Battery capacity [Ah]
E	Endurance [hr]
R	Range [km]
t_{fixed}	Fixed time for each phase [s]
E_{batt}	Usable battery energy [kWh]
LHV	Lower heating value of fuel [Jkg^{-1}]
m_{fuel}	Initial fuel mass [kg]
C_μ	Thrust coefficient
C_p	Power coefficient (Regeneration)
f_s	Spanwise influence factor
f_w	Wing fraction in slipstream
$\dot{I}_m$	Motor current
C_a	Action contribution
S_t	Tracking score
E_h	Hierarchical effectiveness
S_c	Constraint score
V_{wash}	Propwash velocity
DEP	Distributed electric propulsion
MPC	Model predictive control
ECMS	Equivalent consumption minimization strategy
MTOW	Maximum take-off weight
SAF	Sustainable aviation fuel

Greek Letters

η_{prop}	Propeller efficiency [-]
θ, ϕ, ψ	Euler angles [rad]
ω_j	Motor rotor speeds [rad/s]
η_{gen}	Generator efficiency [-]
ϕ_{phase}	Phase factor
η_{hybrid}	Overall hybrid efficiency [-]
α	Angle of attack [°]
β	Sideslip angle [°]
δ_e	Elevator deflection[rad]
δ_a	Aileron deflection[rad]
δ_r	Rudder deflection[rad]
η_{charge}	Battery charging efficiency [-]
η_{dis}	Battery discharge efficiency [-]
γ	Discount factor
λ_t	Hybrid weighting parameter
$\dot{\Omega}$	The rotational speed

Subscripts

temp	Temperature
SOC	State of charge
elec	Electrical
amb	Ambient conditions
rad	Radiation
x, y, z	Cartesian coordinates
m	Moment of a force
turb	Turbine
batt	Battery
eff	Effective velocity over wing

energy management in HEP aircraft is inherently a high-dimensional, nonlinear, and mission-coupled control problem, where early decisions constrain later operation through battery depletion and thermal margin consumption. Effectively exploiting the benefits of HEP, and DEP in particular, therefore requires control strategies capable of continuous power allocation, long-horizon coordination, and phase-dependent constraint management while remaining robust and computationally tractable.

Energy management strategies for hybrid-electric aircraft typically employ either rule-based heuristics or optimization-based formulations. While rule-based approaches offer computational efficiency and high interpretability, they are inherently limited by their reliance on static logic, often resulting in performance gaps compared to theoretical benchmarks and requiring rigorous, mission-specific parameter tuning [6,7]. Conversely, optimization-based methods such as dynamic programming [8], mixed-integer programming [9], and model predictive control [10], explicitly incorporate system constraints to approach or achieve global optimality, thereby providing a more rigorous framework for complex flight regimes. Doff-Sotta et al. [11] implemented a convex model predictive control (MPC) framework for predictive energy management in series and parallel hybrid-electric aircraft propulsion systems, minimizing fuel consumption over predicted flight paths while enforcing constraints on battery state-of-charge, electric motor power, gas turbine operation, and accounting for mass variation and component losses. Detailed results from simulations on regional aircraft showed optimal fuel minimization with robust constraint satisfaction, demonstrating real-time feasibility and significant improvements over non-predictive baselines. Nonetheless, these studies relied on known future flight paths, used simplified loss models, offered limited handling of uncertainties (such as wind and gust), and lacked integration with nonlinear aerodynamics-propulsion coupling. Zhang et al. [12] extended this with nonlinear MPC, explicitly incorporating aerodynamics-propulsion coupling effects in regional jets, yielding reduced block fuel burn and emissions through more realistic modeling. Wang and Koeln [13] proposed a set-based hierarchical model predictive control framework to coordinate multi-timescale dynamics between fuel and electrical systems in hybrid-electric aircraft, using waysets for inter-level coordination, providing theoretical guarantees of constraint satisfaction and safe operation. Detailed simulation resulted in a 16-state powertrain model and demonstrated superior fuel economy, performance enhancements, and reliable operation at system limits compared to single-timescale MPC approaches. Salehpour et al. [14]

worked on a convex multi-objective optimization framework for energy management and component sizing in hybrid fuel cell power systems for more-electric aircraft, minimizing total fuel consumption, polluted gas emissions, and fuel cell size over full flight missions via a tractable convex reformulation. Detailed results showed over 13% fuel savings and improved overall efficiency compared to non-hybrid conventional aircraft, with simultaneous emission reductions. Nevertheless, these methods typically require explicit flight-phase scheduling, incur significant computational cost, or rely on conservative assumptions that limit adaptability during real-time operation, particularly for full-mission scenarios.

More recently, reinforcement learning (RL) has emerged as a promising alternative for aircraft energy management systems (EMS). By learning complex, nonlinear, and state-dependent optimal policies through direct interaction with high-fidelity simulators, RL offers superior adaptability to uncertainties and varying mission conditions compared to traditional model predictive control (MPC) or rule-based methods, without requiring explicit flight-phase scheduling or precise future predictions. Within this domain, several studies have explored RL for hybrid-electric and fuel-cell aircraft. He et al. [15] developed a hierarchical deep RL framework using twin delayed deep deterministic policy gradient (TD3) agents for thrust and power distribution in series hybrid aircraft, achieving faster convergence, robustness to varying conditions, and reduced fuel consumption. Yu et al. [16] applied deep RL to optimize power allocation in turbo-electric hybrid VTOL systems across demanding flight profiles. Yang et al. [19] proposed a Double Deep Q-Network (DDQN)-based strategy for series hybrid electric flying cars, effectively handling dual-mode (ground and air) power demands while reducing computational complexity. A growing body of work has specifically adopted safe reinforcement learning to address critical safety and component longevity constraints in aircraft EMS. For instance, Xiao et al. [17] introduced a safe RL-based energy management strategy for fuel cell hybrid electric aircraft that constrains fuel cell degradation rates during training and optimization. Their approach demonstrated reduced stack degradation, lower hydrogen consumption compared to standard RL, and improved constraint satisfaction. Ni et al. [18] proposed a reinforcement learning framework with discrete actions for energy-optimal flight strategy in high-altitude long-endurance (HALE) solar-powered aircraft, training an agent to select discrete control commands (thrust, angle of attack, bank angle) that maximize net battery energy by balancing solar absorption, gravitational potential, and aerodynamic efficiency over day-night cycles. Detailed simulation results demonstrated 31% improvement in battery energy after one day-night cycle compared to steady-state constant-speed/altitude strategies and 17% over state-machine approaches, enabling extended endurance without additional hardware. Despite these advances, existing safe RL-based EMS frameworks for aircraft still face important limitations. Many rely on simplified system models, exhibit excessive conservatism that degrades performance, struggle with generalization under real-world uncertainties (such as turbulence, payload variations, or sensor noise), and often lack rigorous validation against strict aviation safety and certification requirements [20]. Moreover, the trade-off between strict safety guarantees and energy efficiency remains challenging, frequently resulting in overly cautious policies that limit the full potential of hybrid-electric propulsion.

Full-mission energy management in series hybrid-electric distributed electric propulsion (DEP) aircraft poses distinct challenges due to significantly fluctuating power requirements throughout various flight phases, stringent safety-critical limitations on battery state of charge (SOC), thermal thresholds, and propulsion power, alongside uncertainties arising from mission variations and model discrepancies. Pure MPC is great at short-horizon constrained optimization and robustness, but it often becomes too strict or cautious on long missions and has trouble adapting to new profiles without accurate forecasts. Conversely, standalone Soft Actor-Critic (SAC) reinforcement learning offers strong adaptability, long-term optimization, and generalization in stochastic environments through data-driven learning. Yet, it lacks inherent guarantees for hard constraint satisfaction which are critical for aviation safety. By using a hierarchical MPC-SAC design, we can address these complementary limitations by leveraging the strengths of both: the upper-level SAC agent provides adaptive, mission-aware guidance for strategic decisions across the full mission, while the lower-level MPC ensures real-time constraint enforcement and robustness using the RL outputs as references. This synergy delivers superior fuel efficiency, strict safety compliance, and better generalization compared to baseline MPC or standalone RL, making it particularly suitable for safety-critical hybrid-electric DEP aircraft energy management. These limitations drive the necessity for a control framework that integrates the resilience and constraint sensitivity of model-based techniques with the flexibility and mission-centric optimality of learning-based strategies.

In this work, we propose a hierarchical hybrid control architecture that integrates Model Predictive Control (MPC) with a Soft Actor-Critic (SAC) reinforcement learning agent for efficient energy management and power allocation in a series hybrid-electric DEP aircraft. The SAC agent learns a continuous-action power allocation policy over full mission profiles, while the MPC layer provides conservative, model-based authority during critical operating phases. The two controllers are combined using phase-adaptive blending, with a robust fallback mechanism that ensures safe MPC-only operation in the event of RL failure. The proposed framework is implemented within a high-fidelity simulation environment parameterized using publicly available NASA X-57 data, capturing DEP-specific aerodynamic–propulsive interactions, turbine fuel consumption characteristics, and asymmetric regenerative operation during descent. The key contributions of this paper are as follows:

- **Hierarchical MPC–Reinforcement learning architecture:** Integrates model predictive control for safety-critical phases with reinforcement learning for energy-optimal power allocation in less constrained flight regimes.
- **Decoupled energy management from trajectory control:** Aircraft velocity evolves from thrust–drag dynamics under a simple feedback controller enforcing phase-dependent speed targets, eliminating the need for online constrained propulsion or speed optimization.
- **Phase-adaptive continuous blending and safety fallback:** Controller outputs are blended by flight phase, with automatic reversion to MPC-only operation in case of learning agent degradation, ensuring safe and reliable execution.
- **Full-mission training in distributed electric propulsion environment:** Policies are trained over complete mission profiles using off-policy experience replay in a simulation incorporating DEP-specific aerodynamics, regenerative descent, and constrained fuel maps, achieving energy efficiency, SOC maintenance, and significant CO_2 reduction.

The paper is organized as follows: Section 2 provides a detailed mathematical formulation of the distributed electric propulsion (DEP) aircraft dynamics, asymmetric regeneration and the hybrid-electric powertrain. Section 3 introduces the energy management problem along with the hierarchical energy management strategy, detailing the integration of Soft Actor-Critic (SAC) reinforcement learning with Model Predictive Control (MPC), alongside adaptive time-stepping and emissions modeling. In Section 4, the optimality and real-time feasibility of the model, battery-fuel trade-off, the impact of Sustainable Aviation Fuel (SAF) integration, and the performance of the hybrid MPC-SAC controller are analyzed. Finally, Section 5 summarizes the key findings and suggests avenues for future work.

2. Aircraft and propulsion system modeling

2.1. Flight dynamics of DEP aircraft

The key equations for the nonlinear state-space model for a distributed electric propulsion (DEP) aircraft, incorporating 6-DOF rigid-body dynamics, kinematics, and first-order electric motor/propeller dynamics with propeller wash effects are presented briefly in this section. The full state vector includes angular rates, body velocities, motor current, and rotor speed where $\boldsymbol{x} = [p, q, r, u, v, w, \phi, \theta, \psi, I_m, \Omega]^T$. The control inputs consist of the conventional control surface deflections and the motor voltage is given as $\boldsymbol{u} = [\delta_a, \delta_e, \delta_r, V_m]^T$ Therefore, the nonlinear state-space dynamics are expressed as: $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$. The kinematics of the DEP including total airspeed, angle of attack, and sideslip angle are computed as: $V_a = \sqrt{u^2 + v^2 + w^2}$, $\alpha = \tan^{-1}\left(\frac{w}{u}\right)$ and $\beta = \sin^{-1}\left(\frac{v}{V_a}\right)$. The aerodynamic force coefficients in terms of the body-axis force coefficients, including control surface contributions are given as:

$$C_X = \overline{C}_X(\alpha, \beta) + \Delta C_{X_{\delta_e}}(\delta_e) + \Delta C_{X_{\delta_a}}(\delta_a) \tag{1}$$

$$C_Y = \overline{C}_Y(\alpha, \beta) + \Delta C_{Y_{\delta_r}}(\delta_r) \tag{2}$$

$$C_Z = \overline{C}_Z(\alpha, \beta) + \Delta C_{Z_{\delta_e}}(\delta_e) \tag{3}$$

The aerodynamic moment coefficients including aircraft rolling, pitching, and yawing moment coefficients are:

$$C_l = \overline{C}_l(\alpha, \beta) + \Delta C_{l_{\delta_a}}(\delta_a) + \Delta C_{l_{\delta_r}}(\delta_r) \tag{4}$$

$$C_m = \overline{C}_m(\alpha, \beta) + \Delta C_{m_{\delta_e}}(\delta_e) \tag{5}$$

$$C_n = \overline{C}_n(\alpha, \beta) + \Delta C_{n_{\delta_a}}(\delta_a) + \Delta C_{n_{\delta_r}}(\delta_r) \tag{6}$$

Moreover, translational dynamics considering body-frame forces, including propeller thrust aligned with a thrust incidence angle α_T, are:

$$F_x = \overline{q}SC_X + T\cos(\alpha_T) \tag{7}$$

$$F_y = \overline{q}SC_Y \tag{8}$$

$$F_z = \overline{q}SC_Z - T\sin(\alpha_T) \tag{9}$$

The angular accelerations follow standard rigid-body equations, given as:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \mathbf{I}^{-1}\left(\begin{bmatrix} L \\ M \\ N \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix} \right) \tag{10}$$

where $\mathbf{I}$ denotes the inertia matrix and $[L, M, N]^T$ are the aerodynamic and propulsive moments, respectively. Additionally, the motor dynamics, presented as first-order electric motor and propeller dynamics, are modeled as:

$$\dot{I}_m = \frac{1}{\tau_e}(V_m - K_e\Omega - RI_m) \tag{111}$$

$$\dot{\Omega} = \frac{1}{J}\left(K_t I_m - Q_p(\Omega, V_a)\right) \tag{12}$$

where $\tau_e = L/R$ is electrical time constant, K_e and K_t are back-EMF and torque constants, J is combined rotor and propeller inertia, $Q_p(\Omega, V_a)$ is aerodynamic propeller torque (typically quadratic in Ω and dependent on advance ratio), and thrust $T = C_T\rho\Omega^2D^4$ (or equivalent form, modified for propeller wash effects on local flow). These equations form a complete nonlinear 6-DOF model suitable for simulation, control design, and analysis of DEP aircraft with integrated propulsion effects.

2.2. Distributed electric propulsion (DEP) model

In distributed electric propulsion (DEP) systems for hybrid-electric aircraft, a widely adopted control-oriented propwash modeling approach relies on actuator disk theory, enhanced with flight-phase-specific calibrations. This framework integrates a quasi-steady hybrid of Blade Element Momentum Theory (BEMT) and Lifting Line Theory (LLT) approximations. It applies localized one-dimensional momentum theory (via actuator disks) at individual radial stations along the propeller blades, combined with a strip-theory-inspired uniform slipstream representation and an overall angle-of-attack adjustment [23–25]. For each phase of flight, the model computes key parameters including slipstream velocity, effective airflow across the wing, dynamic pressure modifications, reductions in induced drag, total drag forces, and power demands. Evaluations have shown that this approach yields reliable outcomes, particularly when integrated into real-time nonlinear six-degree-of-freedom simulations for DEP-configured aircraft. According to actuator disk theory, the propwash velocity is computed from the aggregate thrust generated by the propellers, where each propeller has a disk area. From the actuator disk theory, the induced velocity at the disk v_i satisfies $T = 2\rho A v_i(V_\infty + v_i)$. For static conditions ($V_\infty = 0$), assuming n identical propellers that each produce T/n, this reduces to $V_{wash} = v_i = \sqrt{\frac{T}{2\rho A_{disk} n}}$.

For forward flight, the expression is given as:

$$V_{wash} = v_i = \frac{V_\infty}{2}\left(\sqrt{1 + \frac{2T}{\rho A_{disk} n V_\infty^2}} - 1\right) \tag{13}$$

where the induced velocity is computed for each propeller j over $k = 1, \ldots, 20$ radial stations $r_k \in [0.2R, R]$ such that:

$$\lambda_k = \frac{r_k}{R} \tag{14}$$

The effective velocity over the wing (V_{eff}) and angle of attack increment are a weighted average of the freestream velocity and the propwash-augmented velocity, based on the fraction of the wing in the slipstream (f_w);

$$V_{eff} = V_\infty + v_{i,avg}.f_w \tag{15}$$

$$\Delta\alpha = \left(\frac{v_{i,avg}}{V_\infty}\right).f_s.f_w \tag{16}$$

Where $f_s = min(1,\ nD/b)$ is the spanwise influence factor and $v_{i,avg} = 1/n\sum_{j=1}^{n} v_{i,j}$ is the average slipstream velocity. The induced velocity $v_{i,j}$ can be computed as:

$$v_{i,j} = \frac{1}{0.8R}\int_{0.2R}^{R} v_i(r)dr \approx \frac{trapz(r_k, v_{i,k})}{0.8R} \tag{17}$$

Where $v_{i,k} = V_\infty \times a_k = V_\infty\left(1 - \sqrt{1 - C_{T,k}}/2\right)$ and $C_{T,k} = 4a_k(1 - a_k)$. The lift produced by "blowing" air over a wing is non-dimensionalized using the Thrust (or Momentum) Coefficient C_μ:

$$C_\mu = \frac{T}{qS} = \frac{T}{\frac{1}{2}\rho V_{eff}^2 S} \tag{18}$$

$$\Delta C_L = k\sqrt{C_\mu}\left(\frac{f_w}{f_{ref}}\right) \tag{19}$$

The parameter k is an empirical circulation factor, typically ranging from 0.8 to 2.05, which accounts for lift enhancement due to propeller-wing interaction. f_{ref} is a reference coverage (0.76) and represents the maximum achievable wing immersion in the propeller slipstream. In high-lift configurations with multiple active propulsors mode ($n > 2$), k typically varies between 1.5 and 2.05, while in cruise conditions ($n =$

2), it reduces to approximately 0.8 to 0.9. The model captures propwash effects by estimating the slipstream-induced velocity using actuator disk theory and relating the resulting lift augmentation to the momentum coefficient C_μ. By scaling the empirical factor k with the wing slipstream coverage fraction f_w, the formulation represents the variation in lift enhancement across different flight phases, from takeoff to descent. Fig. 1 and Table 1 shows the aircraft and parameters used for the 6-DOF modeling.

2.3. Hybrid powertrain model

The proposed series architecture is shown in Fig. 2. In the series architecture, the propulsion system power output drive P_{drive} is delivered by an electric motor taking electrical power P_{elec} from two sources: a battery with effective power output P_c and a turbogenerator set (gas turbine in series) with power output P_{gen} . The power balance is given as:

$$P_{drv}(t) = P_{elec}(t) = P_{batt}(t) + P_{gen}(t) \tag{20}$$

2.4. Asymmetric regeneration

The asymmetric regeneration feature in the hybrid-electric distributed propulsion (DEP) aircraft simulation calculates the recoverable power from reversed cruise motors during descent, treating them as wind-turbine generators. The core formula is derived from the standard wind turbine power equation [30], adapted for the X-57 Maxwell-inspired configuration with two wingtip cruise motors. Based on Betz's limit and empirical calibration, this uses the wind turbine power equation for extractable kinetic energy from airflow to generate electrical power. It is given as:

$$P_{regen,single} = \frac{1}{2}\rho A_{disk} V_{flow}^3 C_p \eta \tag{21}$$

where C_p is the power coefficient, representing the fraction of airflow energy extractable during reverse-mode operation. In this study, C_p is tuned to 0.38 based on X-57 wind tunnel data, compared to the theoretical betz limit of 0.593. V_{flow} is the effective velocity, and η is the motor/generator efficiency in regeneration mode. The battery charging power, P_{charge} and SOC can be defined as:

$$P_{charge} = \min\left(P_{regen,total}, P_{max,charge}\right) \times \eta_{charge} \tag{22}$$

$$\Delta SOC = min\left(\frac{\Delta E_{regen}}{E_{capacity}}, SOC_{max} - SOC_{current}\right) \tag{23}$$

where $P_{max,charge}$ is the battery maximum charge rate, $\Delta E_{regen} = P_{charge} \times \Delta t \times \eta_{charge}$ $(in\ kWh)$, $E_{capacity}$ is the total battery capacity and Δt is the time step integrated over descent flight phase.

Table 1
Airplane reference parameter of X-57 Mod III/IV [37,38].

Parameter	Value (Metric)	Units
Wing Span	9.642	m
Wing Area (S)	6.194	m^2
Aspect Ratio (AR)	15	-
Mean Aerodynamic Chord (MAC)	0.649	m
Root Chord	0.76	m
Tip Chord	0.53	m
Taper Ratio	~0.70	
Wing Loading (at 1361kg MTOW)	~220	kg/m^2
High-Lift Propellers	12 (along leading edge)	-
Cruise Propellers	2 (at wingtips)	-
Overall Length	~8.7	m
Height	~2.7	m

3. Problem formulation

For a series hybrid-electric aircraft, such as the NASA X-57 Maxwell DEP configuration, the energy management problem involves determining the optimal power split between the battery and turbogenerator at each discrete time step. The objective is to satisfy propulsive power requirements while minimizing total fuel consumption over the entire mission profile. This must be achieved while respecting battery health constraints, actuator limits, and flight-phase-specific operational rules. The proposed hierarchical MPC and SAC framework combines the safety and constraint satisfaction guarantees of Model Predictive Control (MPC) with the adaptability of Soft Actor-Critic (SAC) reinforcement learning.

3.1. System dynamics

3.1.1. Battery SOC dynamics

The battery state-of-charge (SOC) equation describes its evolution from time t to $t+1$ as a function of charging and discharging power. The SOC decreases during discharging based on battery power, time step, discharge efficiency, and battery capacity, and increases during charging or regenerative operation according to the corresponding input power, charge efficiency, and energy capacity. The equation is written as:

$$\mathbf{SOC}_{t+1} = \mathbf{SOC}_t + \begin{cases} \dfrac{\boldsymbol{P}_{\mathrm{batt},t}\,\boldsymbol{\Delta t}}{3600\,\boldsymbol{\eta}_d\left(\mathbf{SOC}_t, \boldsymbol{T}_{\mathrm{batt},t}\right)\boldsymbol{E}_{\mathrm{batt}}}, & \textbf{if } \boldsymbol{P}_{\mathrm{batt},t} > 0 \\ \dfrac{\left(\boldsymbol{P}_{\mathrm{turb},t} + \boldsymbol{P}_{\mathrm{reg},t}\right)\boldsymbol{\eta}_c\,\boldsymbol{\Delta t}}{3600\,\boldsymbol{E}_{\mathrm{batt}}}, & \textbf{if } \boldsymbol{P}_{\mathrm{turb},t} + \boldsymbol{P}_{\mathrm{reg},t} > 0 \end{cases} \tag{24}$$

Tables 2 and 3 shows the parameters set for the battery modelling and the gas turbine and generator models can be found in Refs. [32–35].

3.1.2. Power balance and generator model

The power balance ensures that the required propulsion power is met



Fig. 1. Illustration of the NASA Maxwell-X57 Mod IV aircraft showing DEP configuration [21,22].

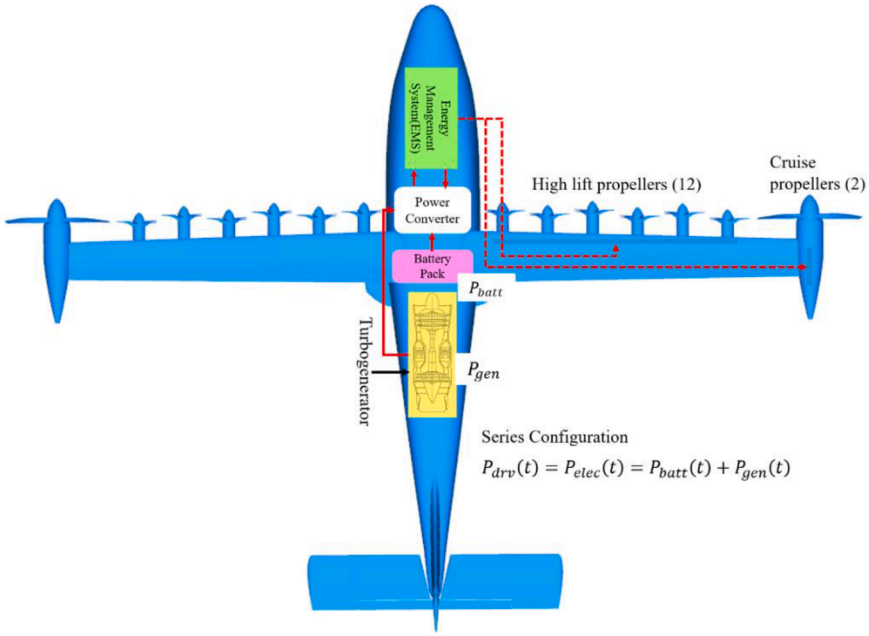


Fig. 2. Illustration of series hybrid-electric propulsion system for NASA X-57 aircraft (not considering aerodynamic stability).

Table 2
Battery parameters for the EP and HEP battery pack.

Category	Parameter	All-Electric X-57 (reference)	Hybrid-Electric X-57	Units
Electrical Parameters (Battery)	Nominal capacity	150	70	Ah
	Nominal energy	69.1	25.6	kWh
	Usable energy	47	16.5	kWh
	Nominal voltage	460	460	V
	Internal resistance	~0.06	~0.06	Ω
	Charge efficiency (target)	0.95	0.95	
	Discharge efficiency (target)	0.95	0.95	
	Max charge current	200	90	A
	Max discharge current	~800	370	A
State of Charge (SOC) Parameters	Initial SOC	1.0	1.0	-
	Minimum SOC	~0.32	~0.32	-
	Maximum SOC	1.0	1.0	-
	Usable SOC window	~68	~68	%
Battery Thermal Parameters	Pack mass	390	200	kg
	Surface area	1.3	0.81	m^2
	Specific heat capacity	1000	1000	J/kg·K
	Total thermal capacity	390,000	182,000	J/K
	Convective heat (still air)	3.25	3.25	W/m^2·K
	Emissivity	0.9	0.9	

by the combined contributions of the battery and the generator. The generator output power is modeled as the turbine power scaled by its efficiency, while regenerative power is assumed to charge the battery directly and does not contribute to propulsion. This is written as:

Table 3
Parameter constraints for the gas turbine engine and generator.

Engine/Generator Parameters	Turboshaft & Generator	Units
Continuous power output	120	kW (shaft)
Peak power output	150	kW (shaft)
Efficiency (fuel-to-electric)	33.3	%
Fuel type & energy density	Jet-A, 12 SAF, 12	kWh/kg
Max fuel capacity	100	kg
Engine mass	57	kg
Generator mass	40	kg
Fuel system mass	30	kg
Operating temperature	80-150	°C (package)
Emissions (CO_2 equiv.)	~0.84	kg/kWh
Total battery energy used	~21.3	kWh (peaks only)
Total system mass (battery & powerplant)	~289	kg
Generator efficiency (η_{gen})	0.95	-
Power time constant (τ)	2	s
Ramp up rate	80	kW/s
Ramp down rate	120	kW/s
SFC coefficients	[0.28, 0.04, 0.02]	kg/kWh

$$P_{\mathrm{req},t} = P_{\mathrm{batt},t} + P_{\mathrm{gen},t} \tag{25}$$

$$P_{\mathrm{gen},t} = \eta_{\mathrm{gen}}\left(P_{\mathrm{turb},t}\right) P_{\mathrm{turb},t} \tag{26}$$

Regenerative power $P_{\mathrm{reg},t}$ is assumed to feed the battery directly and does not contribute to propulsion power. Detailed specifications for the hybrid electric propulsion system architecture are described in full in the previous work of the authors in Ref. [36] and are summarized here in Table 3 for brevity. By referencing this prior documentation for the complete configuration and assumptions, the present work remains focused on its primary analytical findings. All parameters critical to the current methodology are provided herein.

3.1.3. *Gas turbine fuel consumption*

The fuel consumption rate is calculated as the product of the turbine power and its corresponding specific fuel consumption. The total fuel mass is then updated by integrating the fuel flow rate over the time step [31]. This is written as:

$$\dot{m}_{\text{fuel},t} = \text{SFC}\left(P_{\text{turb},t}\right) P_{\text{turb},t} \tag{27}$$

$$m_{\text{fuel},t+1} = m_{\text{fuel},t} + \dot{m}_{\text{fuel},t} \frac{\Delta t}{3600} \tag{28}$$

3.1.4. Finite-horizon optimal control problem (Low-level MPC)

The low-level MPC solves a receding-horizon optimal control problem that minimizes the cumulative stage cost over a prediction horizon N together with a terminal SOC cost. The optimization is subject to battery SOC and fuel dynamics, power and control constraints, SOC bounds, and a terminal SOC requirement to ensure feasible and sustained operation. At time t, the MPC solves the following receding-horizon problem over horizon N:

$$\begin{aligned} \min_{\{\beta_{t+k|t}\}_{k=0}^{N-1}} \quad & \sum_{k=0}^{N-1} J_{t+k|t} + V_f\left(\widehat{SOC}_{t+N|t}\right) \\ \text{s.t. } & \text{SOC and fuel dynamics as defined above} \\ & 0 \le \widehat{\beta}_{t+k|t} \le 1 \\ & \widehat{P}_{batt,t+k|t} \le P_{\text{batt,max}} \\ & \widehat{P}_{turb,t+k|t} \le P_{\text{turb,rated}} \\ & \widehat{SOC}_{\min} \le \widehat{SOC}_{t+k+1|t} \le SOC_{\max} \\ & \widehat{\beta}_{t+k|t} \le \beta_{\max} \\ & \widehat{SOC}_{t+N|t} \ge SOC_{\text{terminal}} \end{aligned} \tag{29}$$

3.1.5. Stage and terminal cost

The stage cost penalizes fuel consumption, deviation of the predicted SOC from its target value, reduced generator efficiency and low SOC operation through a recharge penalty. The terminal cost enforces convergence of the SOC toward the desired target at the end of the horizon, and the first optimal control input is applied as the MPC reference. Mathematically, this is written as:

$$\begin{aligned} J_{t+k|t} \quad &= w_f \dot{m}_{\text{fuel},t+k|t} \frac{\Delta t}{3600} + w_{\text{SOC}} \left(\widehat{\text{SOC}}_{t+k+1|t} \text{SOC}_{\text{target}}\right)^2 \\ &+ w_\eta \frac{1}{\widehat{\eta}_{\text{gen},t+k|t}} + r_{\text{recharge}} \max\left(0, \text{SOC}_{\text{low}} - \widehat{\text{SOC}}_{t+k|t}\right) \end{aligned} \tag{30}$$

And for terminal cost

$$V_f(\text{SOC}_{t+N}) = w_T \max\left(0, \text{SOC}_{\text{target}} - \text{SOC}_{t+N}\right)^2 \tag{31}$$

The first optimal control $\beta^*_{t|t}$ is applied as the MPC reference $\beta_{\text{ref},t}$.

3.1.6. Mission-level global optimization

The mission-level global optimization seeks to minimize the total fuel consumption over the entire mission horizon. This offline problem optimizes the control sequence subject to the full system dynamics and all operational constraints across the mission duration. For offline benchmarking, the full-mission optimal control problem is written as:

$$\min_{\{\beta_t\}_{t=0}^{T-1}} \; m_{\text{fuel},T} \tag{32}$$

subject to the defined dynamics and all operational constraints over the mission horizon T.

3.2. Hierarchical control architecture

The proposed hybrid controller integrates a hierarchical Model Predictive Control (MPC) [26,27] for structured long- and short-term planning with a Soft Actor-Critic (SAC) reinforcement learning agent [28,29] for adaptive real-time decision-making. This architecture leverages the strengths of both approaches: MPC provides constraint satisfaction, recursive feasibility, and optimal handling of forecasted energy profiles, while SAC enables data-driven adaptation to model uncertainties, unmodeled disturbances, and long-term reward maximization through entropy-regularized exploration. Inspired by implementations for NASA X-57 Maxwell-style hybrid-electric distributed propulsion (DEP) aircraft, the framework is enhanced with phase-specific blending weights, PID control for speed regulation, adaptive time-stepping for simulation efficiency, and asymmetric regeneration during descent. The system targets endurance of approximately 0.38 h with optimized MTOW (e.g., 12-20 kWh battery, 10-70 kg fuel). The overall control loop operates as follows: at each timestep t, the environment provides the current state s_t and reward r_t. Both the hierarchical MPC and SAC agent process s_t in parallel. The MPC computes a reference control signal $P_{\text{ref},t}$ based on long-term energy forecasts, while SAC outputs a stochastic action u_t^{SAC}. These are fused in the Hybrid Fusion Module to produce the applied control u_t, which is sent to the environment. Transitions are stored in a prioritized experience buffer to update the SAC policy offline.

3.2.1. System model and environment

The aircraft energy management problem, tailored to hybrid-electric DEP with a series-hybrid configuration (i.e, battery and turbine generator), is modeled as a discrete-time dynamical system:

$$s_{t+1} = f(s_t, u_t, d_t), r_t = \rho(s_t, u_t, s_{t+1}) \tag{33}$$

Where $s_t \in \mathbb{R}^{n_s}$ is the state vector (e.g., position, velocity, altitude, battery state-of-charge (SOC), harvested energy reserves, turbine RPM, battery temperature), $u_t \in \mathbb{R}^{n_u}$ is the control input (e.g., thrust command, power split between propulsion and battery, aerodynamic control surfaces, motor voltages for 14 DEP motors), $d_t \in \mathbb{R}^{n_d}$ represents exogenous inputs, including forecasted harvestable power $P_{\text{harv},\,t:t+N}$ over the prediction horizon, air density variations, and propwash effects. $\rho(\cdot)$ is the reward function, designed to encourage energy efficiency, mission completion, and safe operation, (e.g.$\rho = w_1 \cdot \text{progress} - w_2 \cdot \Delta\text{SOC} - w_3 \cdot \text{constraint violations}$) with additional terms for fuel efficiency and SOC management in implementation. The environment provides perfect observation of s_t, r_t, and a deterministic forecast of harvestable power over the planning horizon. In practice, for X-57-like aircraft, the model incorporates nonlinear DEP dynamics with 14 motors (12 high-lift, 2 cruise), battery thermal models, and turbine SFC lookup table. Fig. 3 shows the flight phase forward velocity (m/s) and altitude (m) for the NASA X-57 aircraft.

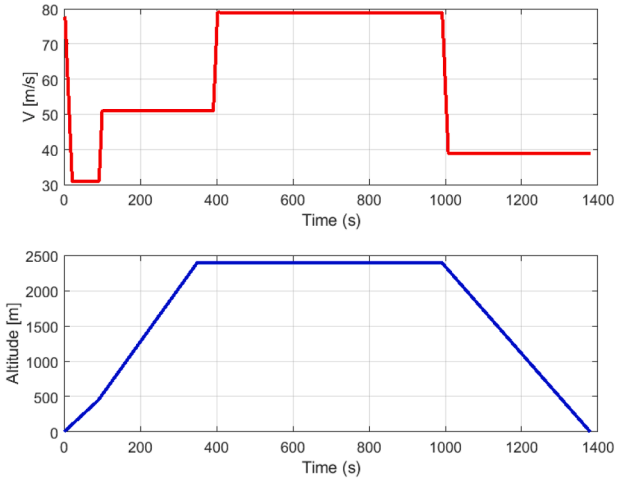


Fig. 3. Plot showing the velocity and altitude profiles at each flight phase of X-57-inspired aircraft.

3.3. Hierarchical MPC layer

3.3.1. High-level MPC (Long-term planning)

The high-level MPC operates at a slower sampling rate (e.g., every T_h seconds) and solves an open-loop optimal control problem over a long horizon N_h(e.g., covering the full mission duration):subject to coarse-grained dynamics:

$$\widehat{s}_{t+k+1|t} = f_{\text{coarse}}(\widehat{s}_{t+k|t}, w_k, \widehat{d}_{t+k|t}), k = 0, \ldots, N_h - 1 \tag{35}$$

And high-level constraints (e.g., total energy balance, waypoint reaching, minimum SOC bounds). Here, w_k are abstracted decision variables (e.g., average power allocation, reference trajectory segments). The terminal cost $V_f(\cdot)$ supports stability and mission completion. The output $w^{\star}_{\text{MPC}}$ serves as a guiding reference for the low-level controller.

3.3.2. Low-level MPC (Short-term execution)

The low-level MPC runs at the control rate and solves a tracking problem over a shorter horizon $N_l \ll N_h$

$$u^{\text{ref}}_{t:t+N_l|t} = \arg\min_{u_{t:t+N_l|t}} \sum_{k=0}^{N_l-1} TBCambriaScriptl_{\text{low}}\left(\widehat{s}_{t+k|t}, u_{t+k|t}\right) + \| \widehat{s}_{t+N_l|t} - s^{\text{ref}}\left(w^{\star}_{\text{MPC}}\right) \|^2_Q \tag{36}$$

Where the weighted norm is defined as $\| x \|^2_Q := x^{\top} Q x$, subject to fine-grained dynamics, actuator constraints, and safety limits. The resulting reference is condensed to a single timestep signal:

$$P_{\text{ref},t} = h\left(u^{ref}_{t|t}, w^{\star}_{MPC}, s_t\right) \tag{37}$$

The model employs an adaptive time step $dt \in [0.1, 2.0]$s with nominal $dt = 1.0s$. The high-level MPC operates over a longer prediction horizon of $N_h = 15$ steps (approximately 15 s lookahead at nominal dt), while the low-level SAC policy uses a single-step horizon ($N_l = 1$). This provides a clear hierarchical separation with $N_l/N_h = 0.067$, satisfying $N_l \ll N_h$. A phase-adaptive blending mechanism (λ_{MPC}, λ_{SAC}) dynamically weights the contributions of both layers according to flight phase (e.g., MPC-dominant in takeoff/climb, SAC-dominant in cruise).These values were selected to balance predictive capability with real-time computational constraints on the target platform.

3.4. Soft actor-critic (SAC)

The SAC agent learns an entropy-maximized stochastic policy for robust exploration and adaptation. The actor parameterizes a Gaussian policy:

$$u^{\text{SAC}}_t \sim \pi_\theta(\cdot|s_t) = \mathcal{N}\left(\mu_\theta(s_t), \Sigma_\theta(s_t)\right) \tag{38}$$

often with a reparameterization trick (and commonly tanh-squashed for bounded actions). Twin Q-critics are trained via clipped double-Q learning:

$$Q_{\phi_i}(s_t, u_t) \leftarrow r_t + \gamma E_{s_{t+1}, u'_{t+1}}\left[\min_{j=1,2} Q_{\phi_j}\left(s_{t+1}, u'_{t+1}\right) - \alpha \log \pi_\theta\left(u'_{t+1}|s_{t+1}\right)\right] \tag{39}$$

The actor maximizes the entropy-augmented objective:

$$\theta \leftarrow \arg\max_\theta \mathbb{E}_{u_t \sim \pi_\theta(\cdot|s_t)}\left[\min_{i=1,2} Q_{\phi_i}(s_t, u_t) + \alpha \mathcal{H}(\pi_\theta(\cdot|s_t))\right] \tag{40}$$

Temperature α is adapted to target a desired entropy level. In the implementation, a pre-trained SAC agent is used, with observations including SOC, power demand, battery temperature, fuel rate, phase ID, previous action, and regeneration efficiency.

3.4. Hybrid fusion module

The final applied control fuses the MPC reference and SAC action:

$$u_t = F\left(P_{\text{ref},t}, u^{\text{SAC}}_t, s_t; \lambda_t\right) \tag{41}$$

In this research work, we employed weighted blending:

$$u_t = (1 - \lambda_t) P_{\text{ref},t} + \lambda_t u^{\text{SAC}}_t \tag{42}$$

Where $\lambda_t \in [0, 1]$. λ_t is a trust parameter that can be fixed, annealed over episodes, or adapted online based on performance metrics (e.g., increasing when SAC consistently outperforms MPC on validation trajectories). Phase-specific weights are used. For example, Takeoff: MPC-dominant (e.g., weights $[0.7,\ 0.3]$ meaning $\lambda_t = 0.3$) and Cruise: SAC-dominant (e.g., weights $[0.2,\ 0.8]$ meaning $\lambda_t = 0.8$). Also, smoothing via exponential averaging and rate limiting were implemented to prevent oscillations. Alternative strategies (for ablation studies) include residual control: $u_t = P_{\text{ref},t} + \Delta u^{\text{SAC}}_t$ and hard switching based on disagreement thresholds.

3.5. Prioritized experience replay and learning

Transitions $\tau_t = (s_t, u_t, r_t, s_{t+1})$ are stored in a prioritized replay buffer B. Priorities are set proportional to temporal difference (TD) error augmented with an MPC consistency term:

$$p_j = \left|\delta^{\text{TD}}_j\right| + \beta \| u_j - P_{\text{ref},j} \| + \epsilon \tag{43}$$

to emphasize transitions where SAC deviates significantly from MPC (potential learning opportunities) or exhibits high temporal-difference error. SAC parameters are updated off-policy using prioritized mini-batches. In practice, the agent is pre-trained offline. It is worth mentioning that, the phase-specific blending coefficients were selected based on three key factors: operational risk, constraint tightness, and model reliability in each flight phase. Safety-critical phases with high thrust demand, rapidly changing dynamics, and low tolerance for constraint violations (e.g., power limits, SOC bounds, actuator saturation) favor strong MPC dominance to leverage its explicit constraint enforcement and deterministic safety guarantees. In contrast, steady-state cruise features looser constraints, more stable airspeed/altitude, and a primary goal of long-term efficiency optimization, allowing greater authority to SAC RL, whose entropy-regularized policy supports better exploration and adaptation to minor modeling uncertainties. This rationale is summarized in Table 5. Mathematically, the MPC weight can be conceptually expressed as proportional to the combined influence of these factors:

$$w^{\star}_{\text{MPC}} \propto \frac{\textit{Constraint Tightness} \times \textit{Time Criticality}}{\textit{Model Uncertainty}} \tag{44}$$

Table 5
Phase-specific weighting for hierarchical MPC and SAC RL control.

Flight Phase	MPC Weight	SAC RL Weight	Rationale
Global Default	0.7	0.3	Fallback weights; MPC-dominant for safety if phase information is unavailable.
Takeoff / Initial Climb	0.9	0.1	Safety-critical; MPC dominates to enforce constraints during high power demand
Climb Transition	0.8	0.2	MPC still dominant; slight SAC authority for efficiency optimization.
Cruise	0.3	0.7	Loose constraints; SAC RL dominates to optimize fuel efficiency.
Descent / Approach / Landing	0.6	0.4	Balanced; MPC ensures stability and SOC management, SAC provides fine-tuning.

(Note: The actual numerical values were not derived via direct optimization of this formula but served as a guiding heuristic). These preliminary ratios were initially inspired by established practices in hierarchical MPC-RL fusion [43,44] for safety-critical systems, where MPC is typically assigned higher authority for hard constraint handling and RL augments performance in less constrained regimes. The final weights were then refined iteratively through extensive closed-loop simulations in high-fidelity environments. In each iteration, we evaluated performance metrics such as constraint violation rate, fuel/energy consumption, trajectory tracking error, and passenger comfort (e.g., vertical acceleration). The coefficients were adjusted in increments of 0.1 until a satisfactory safety-efficiency trade-off was consistently achieved across multiple mission profiles. Here, flight phases are primarily predefined based on standard mission profiles and real-time flight parameters such as altitude, airspeed, vertical speed, and thrust setting. Phase detection uses simple threshold-based logic. To address potential ambiguity in phase boundaries caused by uncertainty, noise, or off-nominal missions, the framework automatically switches to the Global Default weights (MPC: 0.7, SAC RL: 0.3). This conservative fallback ensures MPC dominance for safety and constraint enforcement whenever phase identification is unreliable, providing robust performance without requiring complex online phase inference.

In summary, the algorithm for the hybrid MPC-SAC is written as shown in Table 4 and the flow chart for the proposed model is presented in Fig. 4;

The training progress of the Soft Actor-Critic (SAC) agent, as illustrated in Fig. 5, demonstrates a robust and efficient learning trajectory across 1,400 episodes. During the initial phase (episodes 0-150), the agent maintains a relatively stable, low reward as it prioritizes exploration, a behavior driven by SAC's maximum entropy objective which prevents premature convergence to sub-optimal policies. A significant performance breakthrough occurs between episodes 150 and 300, characterized by a sharp, monotonic increase in episode rewards. Following this rapid acquisition phase, the agent enters a refinement stage where rewards plateau around a mean value of 85. The narrow variance between the raw episode data (light blue) and the moving average (dark blue) indicates a stable convergence, while the maintained stochasticity in the later episodes confirms that the agent has successfully optimized the trade-off between exploration and exploitation. Overall, the absence of catastrophic forgetting or significant oscillations suggests that the hyperparameters were well-tuned for this environment, resulting in a highly reliable and sample-efficient policy.

Table 4
Overall algorithm.

Algorithm 1: Hybrid Hierarchical MPC-SAC Execution and Training

Initialize SAC networks (actor θ, critics φ_1, φ_2), replay buffer B, λ_0
Initialize MPC solvers (high-level and low-level)
for each episode **do**
  Reset environment → s_0
  for each timestep t **do**
    Observe s_t and forecast $P_{\text{harv},t:t+N}$
    Compute w^*_{MPC}(high-level planning)
    Compute $P_{ref,t}$ (tracking / execution MPC)
    Sample $u_t^{SAC} \sim \pi\theta(\cdot|s_t)$
    Fuse: $u_t = (1 - \lambda t)P_{ref,t} + \lambda_t u_t^{SAC}$(phase-weighted)
    Execute u_t, observe s_{t+1}, r_t
    Store τ_t in B with priority
    if training step **then**
      Sample minibatch from B
        Update SAC critics, actor, and α
        Optionally update λ_t based on advantage
    end if
  end for
end for

4. Results and discussion

This section presents a comprehensive evaluation of the proposed hierarchical MPC-guided reinforcement learning (RL) framework for hybrid-electric distributed propulsion aircraft. To assess its performance, the proposed strategy is benchmarked against a standalone MPC approach through high-fidelity mission simulations. The pure MPC strategy serves as a baseline for the 'best possible' performance due to its inherent optimization capabilities and shared system models and constraints. These simulations evaluate performance across several key metrics: fuel efficiency, state-of-charge (SoC) regulation, constraint satisfaction, and robustness against system uncertainties. By comparing these approaches, we quantify the specific impact of integrating the RL module into the control architecture and demonstrate the feasibility of the hierarchical approach for real-time energy management.

4.1. Optimality trade-offs and real-time feasibility

This section discusses the trade-offs between control optimality and real-time feasibility in the proposed hierarchical framework. The influence of MPC prediction horizon length on computational burden is examined, along with the balance between policy optimality and robustness introduced by reinforcement learning guidance.

In Fig. 6, the relative performance of the model predictive control method (MPC), the pure Soft Actor-Critic algorithm (SAC), and the hierarchical model predictive control-guided SAC algorithm introduced is shown in the case of energy management for a hybrid-electric distributed propulsion system during a simulation with a mission time of 0.38 h. In the top-left graph, the relative effectiveness of the SAC algorithm with respect to the MPC algorithm (denoted as SAC-MPC) is presented. This plot shows strong fluctuations of the relative effectiveness around the value zero with considerable positive and negative excursions (of about +0.9 and -0.5) at time t, within the range of 300 to 1000 s. The SAC deviation from MPC remains bounded throughout the mission, indicating that the learning policy does not merely replicate the MPC solution. Instead, SAC introduces deliberate, context-dependent action modifications, particularly during steady operating phases, reflecting its ability to incorporate long-horizon considerations while remaining effectively constrained by MPC guidance. The deviation remains asymmetric but bounded, suggesting SAC explores alternative energy-splitting strategies while remaining constrained by the hierarchical framework. The top-right subplot shows the comparison in the power split fraction (share of battery vs. generator power) between the three approaches: the setpoints in MPC are clearly stable and piecewise-constant with particular transition instances at 350 and 950 sec (blue line), and as expected, the pure SAC exhibits significant chattering, with commands oscillating across the full control range throughout the entire mission duration (indicated by the red line). In contrast, the proposed framework leverages MPC guidance to produce smooth control actions (green line). This integration closely tracks the reference trajectory while successfully mitigating the chattering behavior observed in the standalone agent. The bottom-left subplot represents the hierarchical effectiveness measure related to the temporal process in transferring the guidance from the MPC to the SAC RL agent. E_h is defined as the hierarchical effectiveness (0-1). Here, 0 and 1 represent poor and excellent effectiveness respectively.

$$E_h = \max(\,0, \min(1,\; w_a C_a + w_t S_t + w_c S_c)) \tag{45}$$

The action contribution (C_a) is defined as:

$$C_a = \frac{1}{N}\sum_{i=1}^{N} \frac{|hybrid_i - MPC_i|}{|SAC_i - MPC_i| + \epsilon} \tag{46}$$

Where $\epsilon = 10^{-8}$ to avoid division by zero. Tracking score (S_t) is also estimated as:

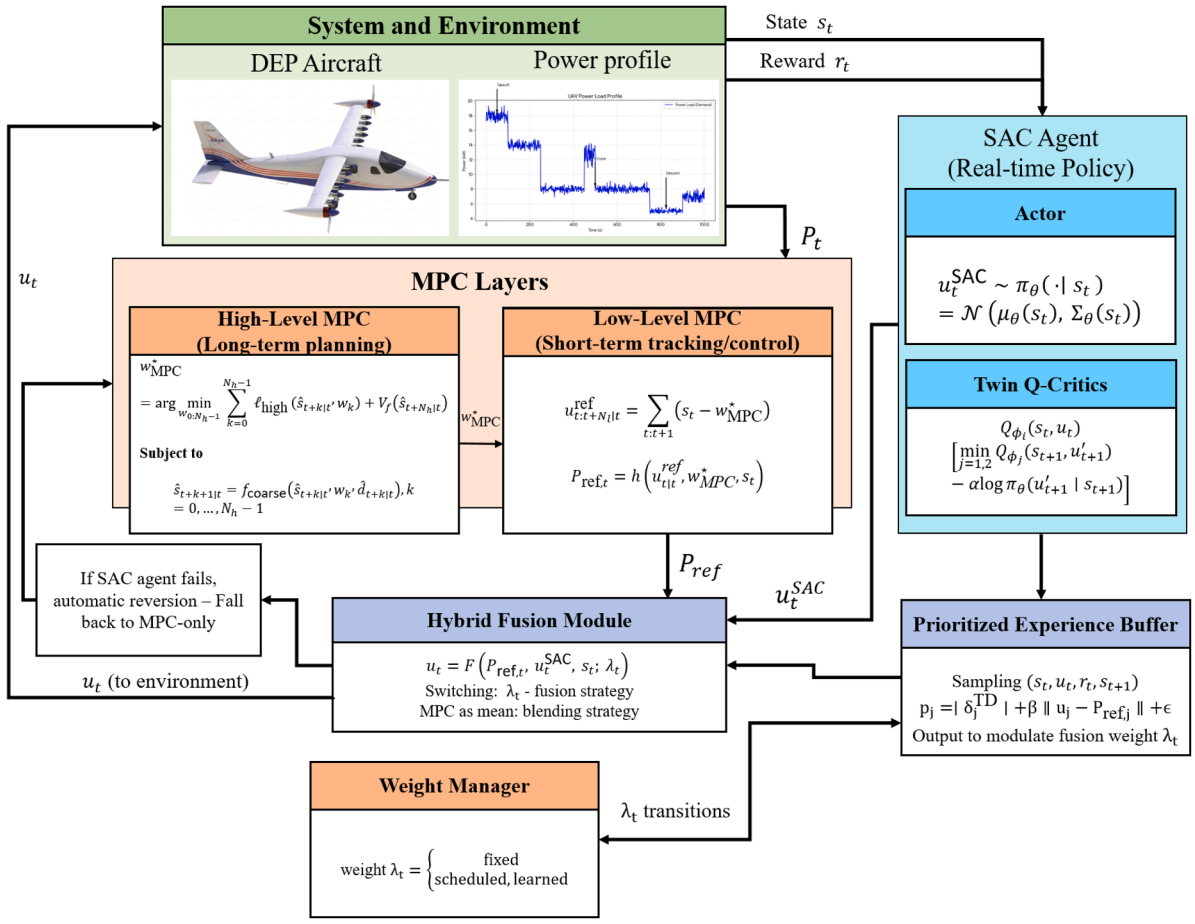


Fig. 4. System architecture for the hierarchical MPC-SAC aircraft energy management strategy.

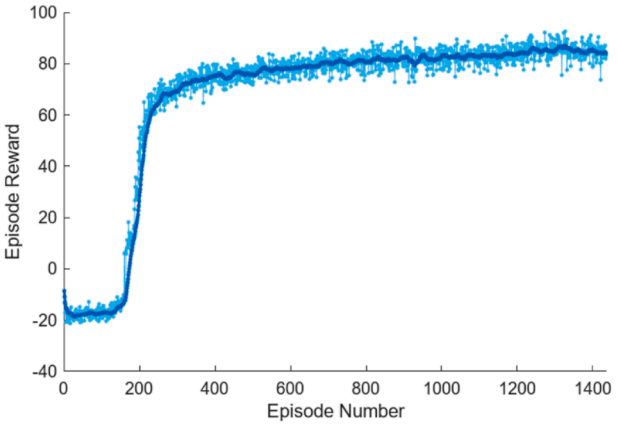


Fig. 5. Cumulative training episode rewards during the training phase, featuring raw data points (light blue) and a smoothed moving average (dark blue line).

$$S_t = max\left(0,\ 1 - \frac{RMSE_{speed}}{V_{cruise}}\right) \tag{47}$$

Constraint score (S_c) is defined as:

$$S_c = max\left(0,\ 1 - \frac{number\ of\ SOC\ violations}{N}\right) \tag{48}$$

C_a measures how much the SAC RL policy (red) is actually changing the final hybrid command (green) compared to pure MPC (blue). Higher value means SAC is contributing meaningfully. S_t shows how well the aircraft follows the target speed. Perfect tracking gives 1.0. S_c shows how well the battery SOC stays within safe limits. Fewer violations equal higher score. N is the total number of time steps (data points) considered in the calculation. Moreover, the weights were defined as $w_a = 0.35$, $w_t = 0.40$, $w_c = 0.25$. These weights were determined heuristically (i.e., based on engineering judgment and practical experience with hierarchical control systems). In aerospace applications involving hybrid-electric propulsion, tracking performance is typically given higher priority to ensure flight stability and energy efficiency, while still allocating meaningful weight to the adaptive contribution of the reinforcement learning layer and constraint satisfaction for battery safety. This approach is widely accepted in the early development and validation stages of hierarchical MPC-RL frameworks, where domain-specific priorities guide initial tuning [39,40]. Subsequent refinement can be achieved through sensitivity analysis or multi-objective optimization once comprehensive performance data across multiple mission scenarios become available. It is evident in the Fig. 6c (left bottom) that the effectiveness initially rises during climb from about 0.4 to almost 0.8. the MPC is seen to strongly control the power split during the safety-critical stage. It consistently exceeds the 0.7 good threshold throughout the cruise phase, reaching near-perfect values of approximately 0.98 between 600 and 900 s. This indicates strong synergistic contribution from the SAC RL layer to the final hybrid power split. A moderate decline is observed during descent, followed by partial recovery toward the end of the mission. The hierarchical controller maintains an overall average effectiveness of 0.82, effectively blending the conservative, constraint-aware nature of MPC with the adaptive optimization capabilities of SAC RL. Lastly, the bottom-right bar chart provides an overview of the essential performance indicators on a

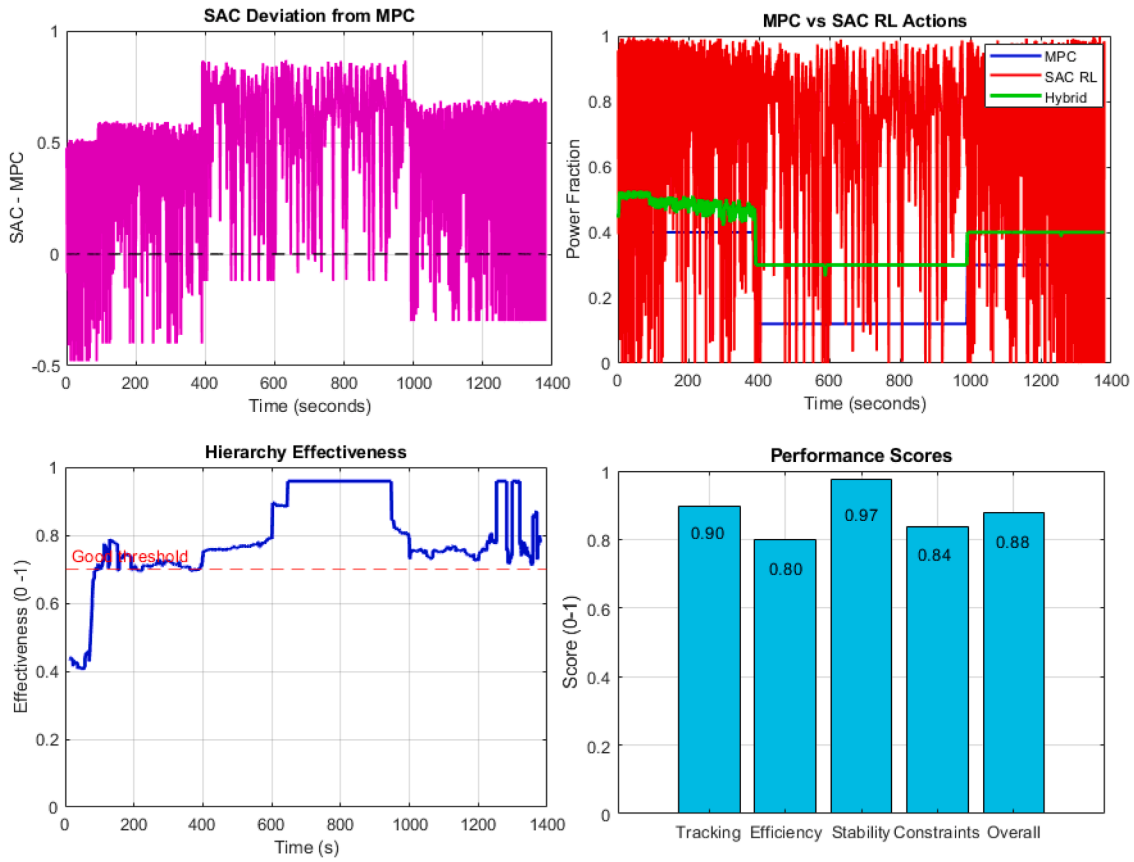


Fig. 6. Comparative performance of MPC, pure SAC, and hierarchical MPC-Guided SAC for hybrid-electric aircraft energy management.

normalized scale from 0 to 1. The hierarchical strategy achieves the highest performance overall, attaining top scores in tracking (0.90) and efficiency (0.80). It slightly outperforms MPC in tracking while maintaining strong constraint satisfaction (0.84) and excellent stability (0.97). As a result, it delivers the highest overall score (0.88), surpassing both the MPC baseline and pure SAC, the latter performing notably worse across all evaluated metrics. The experiments clearly emphasize the fact that the pure SAC algorithm, being highly data-adaptive, demonstrates uncontrolled chattering in the control output and unsatisfactory levels of performance in the safety-critical aeronautical domain. In turn, the hierarchical MPC-SAC framework developed in the current study provides a promising route toward achieving optimal levels of MPC in conjunction with optimal smoothness in the control action in real-time energy management hybrid-electric propulsion systems.

Fig. 7 presents two key aspects of the system performance under the proposed hierarchical MPC-guided reinforcement learning approach for energy management in a hybrid-electric distributed propulsion aircraft over the 1400-second mission. These are the control effort (rate of change in power fraction) and the battery state of charge (SOC) trajectory with respect to its operational constraints. The left subplot shows the control effort, defined as the absolute change in power fraction between consecutive time steps. The trace exhibits low activity during most of the simulation, with small gradual increases in the first ~200 s, followed by a sharp peak reaching approximately 0.17 around 400 s - coinciding with a major transient in the power demand or operating condition. A secondary, smaller spike of about 0.10 occurs near 1000 s, after which the control effort returns to near-zero values for the remainder of the mission. This profile demonstrates that the hierarchical controller produces smooth and sparse control adjustments, with significant effort concentrated only at critical transients, thereby minimizing actuator wear, power electronics stress, and unnecessary energy losses compared to noisier pure RL policies. The right subplot depicts the battery SOC trajectory (solid blue line) alongside the upper (dashed red, ~92-95%) and lower (dashed red, ~20%) safety constraints. Starting at 100%, the SOC decreases steadily to a minimum of approximately 56% around 400 s, reflecting active battery discharge during high-power demand phases. It then gradually recovers to around 65–68% by the end of the simulation through controlled recharging. More importantly, the SOC remains well within the predefined bounds throughout the entire mission, never approaching or violating either the lower 20% threshold (which protects against deep discharge and battery degradation) or the upper limit (which prevents overcharging). This behavior confirms robust constraint satisfaction, a fundamental requirement for safe, reliable operation in aerospace applications. These results underscore the effectiveness of the hierarchical MPC-guided SAC framework in delivering low-effort, smooth control inputs while strictly enforcing battery SOC constraints.

Fig. 8 presents a comprehensive analysis of the hierarchical interactions between the model predictive control (MPC) guidance and the soft actor-critic (SAC) reinforcement learning policy in the proposed framework. The top-left scatter plot displays SAC corrections (SAC − MPC) color-coded by flight phase (takeoff: red; climb: green; cruise: blue; descent: magenta), revealing large and frequent deviations during dynamic takeoff and climb phases, with corrections diminishing markedly in magnitude and scatter during extended cruise, indicating successful imitation learning and reduced need for guidance in steady-state operation. The top-right histogram of correction magnitudes shows a unimodal distribution peaked around 0.4-0.5 with a long tail toward zero and rare large deviations (> 0.8), confirming that hierarchical oversight effectively bounds RL errors to moderate levels. The bottom-left time-series highlights re-planning events triggered by deviations exceeding 10%, with dense activations confined to takeoff, climb, and descent while virtually absent during cruise. This demonstrates phase-dependent intervention that minimizes computational overhead.

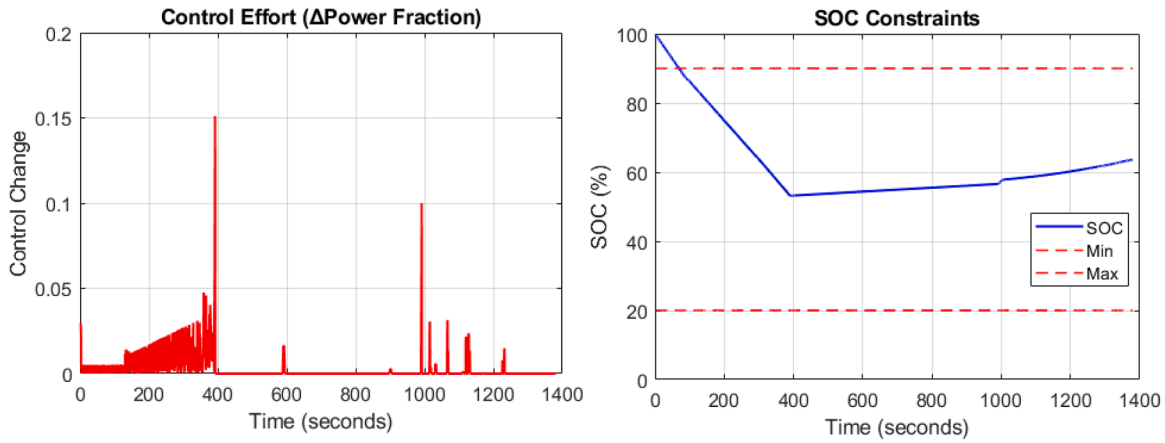


Fig. 7. Control effort and battery SOC trajectory with constraints under hierarchical MPC-Guided SAC approach.

Finally, the bottom-right plot illustrates dynamic control authority blending, starting with dominant MPC weight (~80-90%) early in the mission, followed by sharp transitions around 400 s and 1000 s that progressively shift majority authority (~60–70%) to the faster SAC policy. Collectively, these results underscore the hierarchical framework's ability to ensure safe, near-optimal performance through intensive early supervision and knowledge transfer, enabling autonomous RL operation during cruise with sparse re-planning. This substantially reduces online computational demands while preserving constraint satisfaction and efficiency critical for real-time aerospace applications.

Fig. 9 demonstrates the adaptive time step size employed throughout the mission, showcasing the framework's ability to optimize computational efficiency while preserving accuracy and safety. Starting at a baseline of 0.1 s during initialization, the step size increases progressively in a stepwise manner, reaching intermediate values around 0.5-0.6 s by 200-300s. A sharp escalation to the maximum of 2.0 s occurs near the major transient at 350-400 s, followed by a prolonged plateau at this large value from approximately 450 to 1000 s, coinciding with the most stable cruise phase and aligning closely with near-zero hierarchical guidance effectiveness. At the secondary transient around 1000 s, the step size drops abruptly to 1.0 s and continues decreasing to small values (0.1-0.25 s) toward the mission end, ensuring high-fidelity resolution during critical phases. This dynamic adaptation achieves up to a 20-fold reduction in control iterations during steady-state segments, significantly lowering real-time computational burden for onboard implementation, while automatically reverting to finer steps at transients to maintain numerical stability, precise tracking, and strict constraint satisfaction-key enablers for practical deployment in hybrid-electric distributed propulsion aircraft.

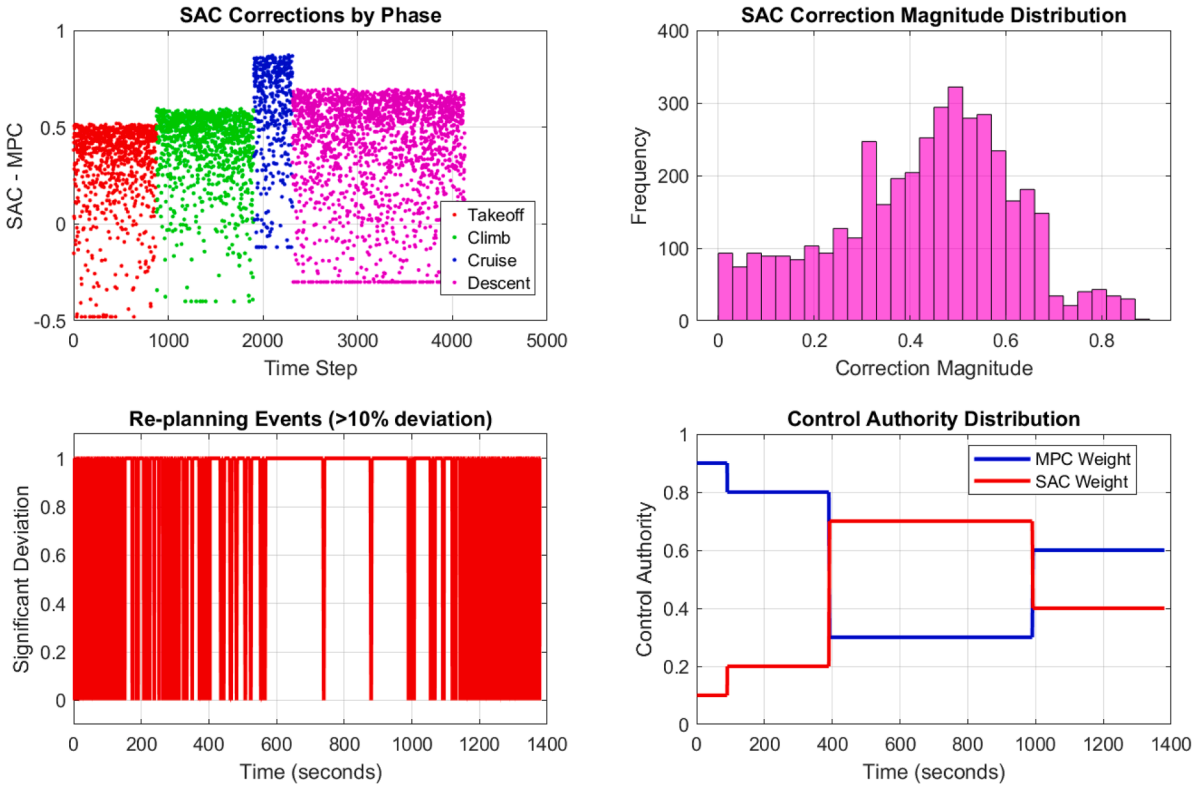


Fig. 8. Hierarchical interactions between the MPC guidance and SAC reinforcement learning policy.

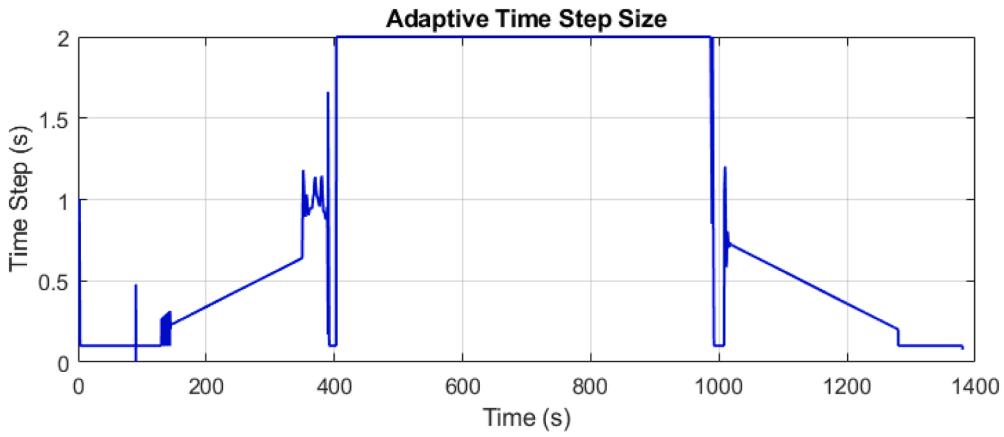


Fig. 9. Adaptive temporal discretization across mission phases under hierarchical MPC-guided reinforcement learning.

4.2. Hybrid MPC-SAC energy management performance

This section assesses the energy management performance of the proposed hybrid MPC–SAC framework for a hybrid-electric aircraft. The results demonstrate how the integration of reinforcement learning enhances power-split decisions and long-horizon energy utilization while maintaining the safety and constraint-handling properties of MPC across mission phases.

Fig. 10 shows a plot of power flows for the generator and battery power source for the complete mission profile. During the initial high-demand phase (0-400 s, corresponding to takeoff and climb), both the generator and battery contribute substantially, with P_{gen} fluctuating noisily around 7-8 $\times$ 10^4 W and $P_{batt,out}$ providing comparable supplementary power (also noisy, peaking near 8 $\times$ 10^4 W), while charging power remains near zero. This combined contribution meets elevated thrust requirements while the battery assists to maintain generator operation near efficient setpoints. At the major transition around 400s (entry into cruise), P_{gen} steps up smoothly to a higher constant level of approximately 10.5 $\times$ 10^4 W, $P_{batt,out}$ drops sharply to near zero, and charging remains negligible, indicating sole reliance on the generator during the long quasi-steady cruise segment for optimal fuel efficiency. Near 1000s (onset of descent or reduced power demand), P_{gen} decreases to a lower steady value of ~4 $\times$ 10^4 W, $P_{batt,out}$ remains minimal, and P_{charge} rises positively to ~8 $\times$ 10^4 W before gradually tapering to ~6 $\times$ 10^4 W, reflecting opportunistic battery recharging using excess generator capacity during low-demand conditions. The minor early-phase noise in generator and discharge traces is effectively suppressed compared to pure RL approaches (as shown in prior figures), highlighting the hierarchical guidance's role in producing smooth, actuator-friendly profiles. Overall, these power flows confirm the strategy's effectiveness in intelligently allocating energy sources. The battery is discharged to support high-power transients, while the generator operates at efficient levels during cruise. During low-demand phases, the battery state of charge is recovered through regenerative charging. Together, these behaviors enhance fuel efficiency, reduce emissions, and ensure robust constraint satisfaction, supporting sustainable hybrid-electric aviation.

Fig. 11 compares the required propulsion power demand ($P_{required}$, solid black) with the actual supplied power ($P_{supplied}$, dashed red) delivered by the hybrid-electric powertrain, demonstrating excellent tracking performance of the hierarchical controller across mission phases. The demand profile begins at a high ~14.5 $\times$ 10^4 W during takeoff (0-100 s), steps down progressively to ~11 $\times$ 10^4 W through climb and cruise entry (~400 s), maintains this level during extended cruise (400–1000 s), then exhibits a sharp drop to ~6 $\times$ 10^4 W at ~1000 s. This marks the onset of descent, followed by a gradual taper toward near-zero by mission end. The supplied power tracks the demand with remarkable precision, showing near-perfect overlap in steady segments and only minor, rapidly corrected transient deviations at step changes. The pronounced sharp drop in both demand and supply around 1000s is attributable to the transition into descent, where thrust requirements decrease substantially and asymmetric regenerative propulsion becomes active: the distributed propellers operate in wind milling mode,

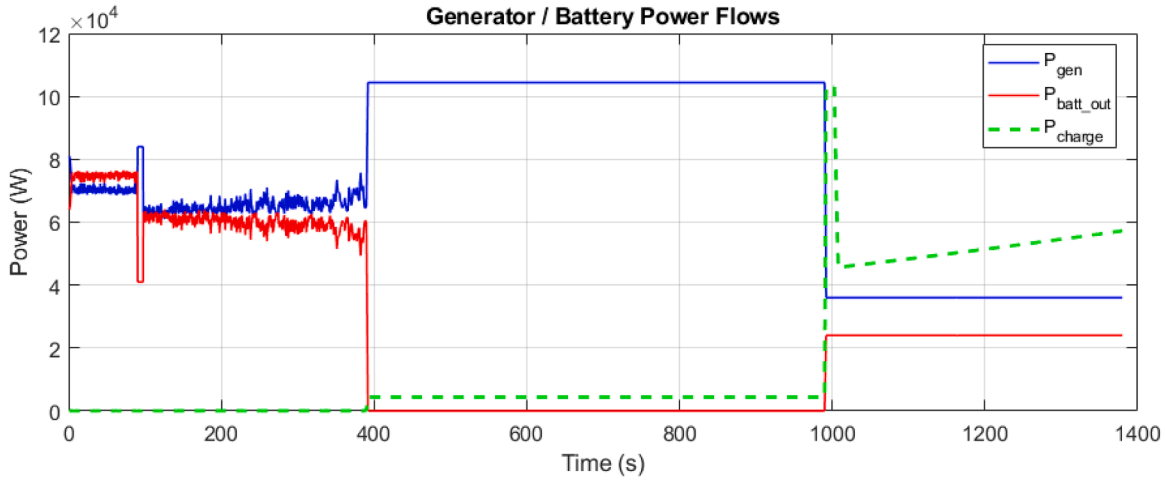


Fig. 10. Time histories of generator power, battery discharge, and charging under hierarchical MPC-Guided SAC control.

generating excess power that is directed toward battery charging (as evidenced by the positive P_{charge} in related power flow figures) rather than requiring symmetric boosting. This opportunistic regeneration during low/thrust-negative phases enables partial state-of-charge recovery without compromising tracking accuracy. Overall, the tight demand-supply balance validates the hierarchical framework's capability to deliver precise, responsive power allocation. This prevents thrust deficits or energy waste while intelligently exploiting asymmetric regeneration for improved efficiency and sustainability in hybrid-electric distributed propulsion aircraft.

This is very evident in Fig. 12, which shows a dual-panel figure detailing the battery utilization and in-flight recharging behavior, complementing the power demand-supply balance shown in the previous figure. The top panel presents the battery SOC trajectory (blue), starting at 100% and decreasing steadily to a minimum of about 55% around 400 s due to intensive battery discharge supporting high propulsion demands during take-off and climb. SOC then remains nearly constant (with slight recovery) through cruise (400 -1000 s), before increasing progressively to about 65% by mission end, driven by opportunistic in-flight recharging. Red markers highlight a single extended charging event during descent (post-1000 s), delivering a total charge energy of 9.79 kWh at a maximum power of 103.0 kW. The bottom panel contrasts battery discharging power (blue, peaking noisily at approximately 70–80 kW early before dropping to near-zero) against charging power (red, negligible until the sharp rise to about 100 kW around 1000 s, followed by a gradual taper to about 60 kW). This charging profile directly correlates with the abrupt power demand reduction observed in the prior figure at ~1000 s, where the transition to descent enables asymmetric regenerative propulsion: distributed propellers windmill to generate excess electrical power, which is intelligently routed to recharge the battery rather than dissipated or symmetrically boosted. The hierarchical controller exploits this low-demand phase for substantial SOC recovery without compromising thrust tracking, demonstrating effective in-flight energy recuperation that enhances overall mission efficiency, reduces fuel consumption, and supports sustainable operation in hybrid-electric distributed propulsion aircraft.

To assess each component's performance, the battery and turbine temperature and efficiencies are presented in detail. Fig. 13 shows a dual-plot figure that characterizes the thermal and efficiency behavior of the lithium-ion battery pack in the hybrid-electric propulsion system, reflecting the thermodynamic and electrochemical responses to the phase-dependent power cycling imposed by the hierarchical controller. The left plot shows the battery bulk temperature trajectory, initiating at ~22°C (ambient/pre-mission equilibrium) and exhibiting a monotonic increase driven by cumulative Joule heating (I^2R losses) and entropic heat contributions during high-rate discharge. The temperature rise is initially gradual (~0.05°C/s average rate from 0-400 s) as the battery supports peak power demands in takeoff/climb with moderate C-rates, then accelerates (steeper slope post around 400 s) to exceed 50°C by mission end, approaching typical upper operational limits for many aerospace-grade cells. This profile indicates sustained internal heat generation outpacing convective/conductive cooling, with no active thermal runaway but highlighting the need for robust thermal management in prolonged high-load missions. The right panel depicts round-trip efficiency components: discharge efficiency $\eta_{discharge}$(blue) and charge efficiency η_{charge}(green). Discharge efficiency begins at ~91% (near peak for low-C-rate conditions), briefly rises due to favorable entropic effects at moderate temperatures, then decays progressively to ~80% as rising temperature exacerbates overpotential losses (activation, ohmic, and concentration polarization) and reduces effective capacity via Arrhenius-dependent reaction kinetics. Charge efficiency starts lower (~85%) owing to higher overpotentials during early low-SOC acceptance, peaks at ~87% mid-mission during limited charging opportunities, then drops sharply post-1000 s to ~76% and plateaus low-consistent with elevated temperatures increasing parasitic side reactions and lithium plating risk during the extended regenerative charging phase in descent (asymmetric regeneration). These efficiency trends quantify the electrochemical penalties of aggressive cycling and thermal excursion, yet remain within acceptable bounds. It also demonstrates that the hierarchical framework has the ability to balance performance optimization against battery health degradation in constraint-satisfied operation.

Similarly, Fig. 14 illustrates the closed-loop dynamics of turbine specific fuel consumption (SFC) and efficiency for the proposed hierarchical MPC-SAC energy management strategy. As expected, the SFC dynamics are bounded and piecewise smooth, confirming stable fuel flow and turbine shaft power control and regulation under high-level supervision. During the period of approximately 400 s to 1000 s, a converged steady-state operation is realized and maintained by the control algorithm around a neighborhood that is characterized by low SFC and high efficiency, defining a locally optimal steady-state equilibrium for the overall propulsion-energy system dynamics. This trend indicates that the high-level MPC control is capable of steering the system state towards desirable operation regions within a finite prediction horizon, supported by the long-term value function shaping of the SAC policy, which favors operation near thermodynamically optimal regions. At other times, particularly during periods of low propulsion demand, the system deliberately operates in less optimal regions,

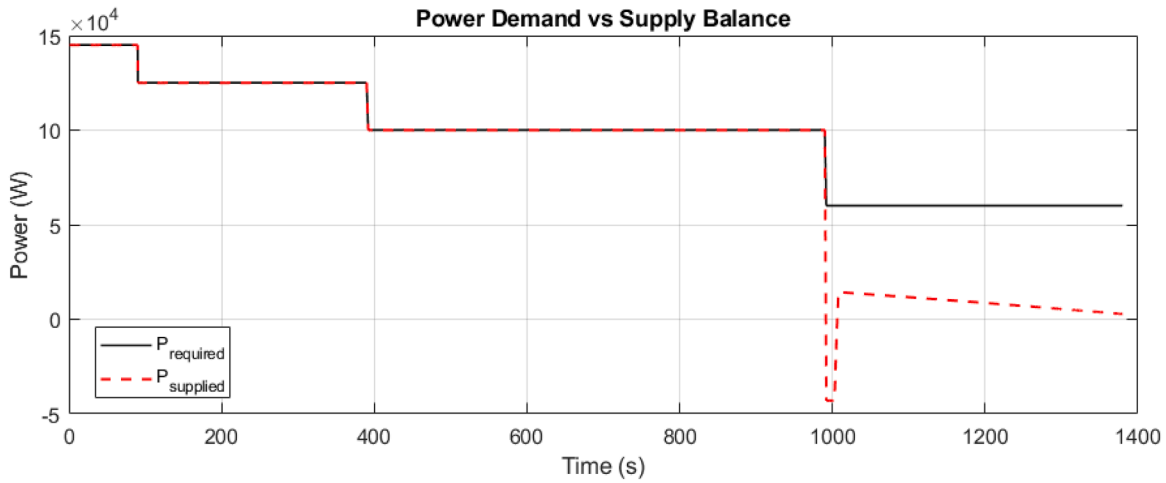


Fig. 11. Power demand versus supplied power balance under the proposed hierarchical MPC-guided SAC energy management strategy.

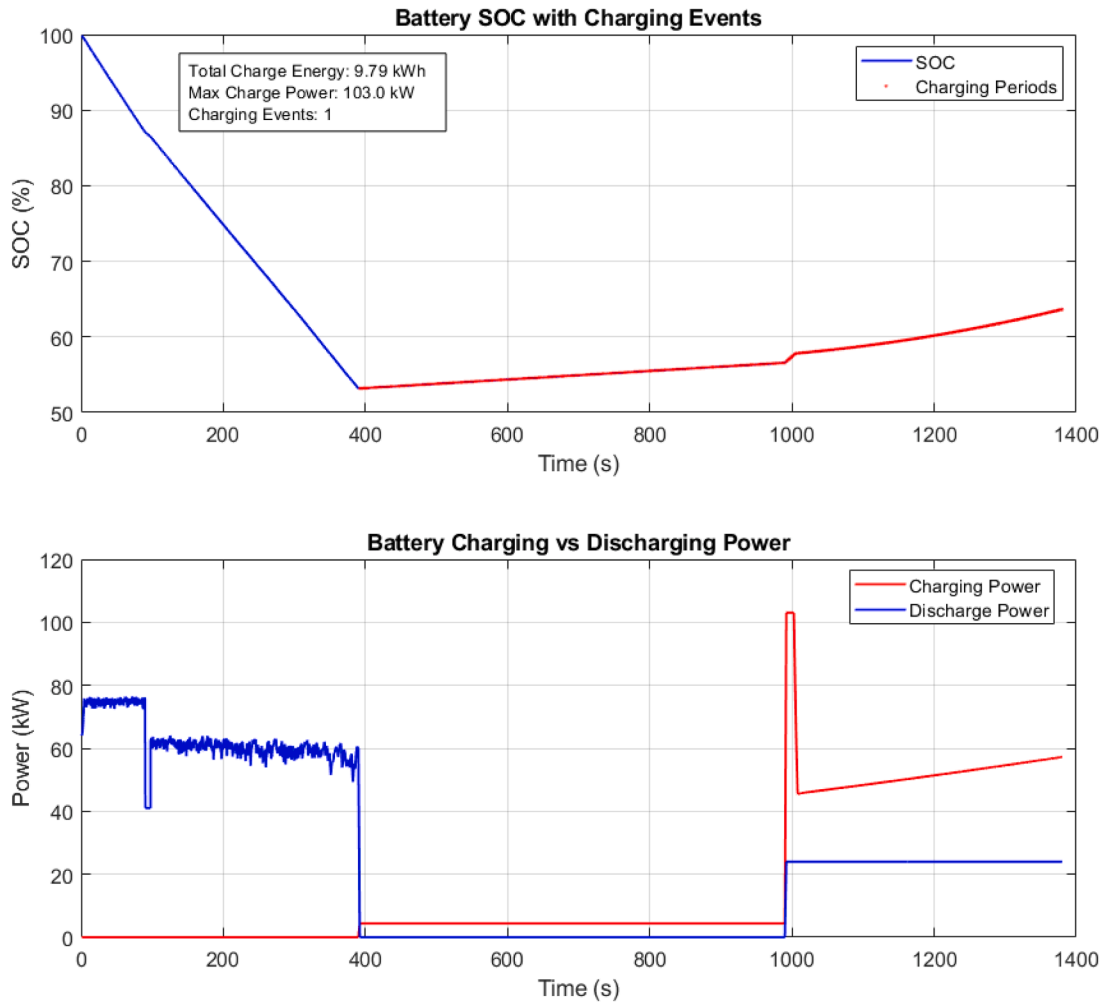


Fig. 12. Battery state-of-charge (SOC) trajectory with in-flight charging and asymmetric regeneration events under the proposed hierarchical MPC-guided SAC energy management strategy.

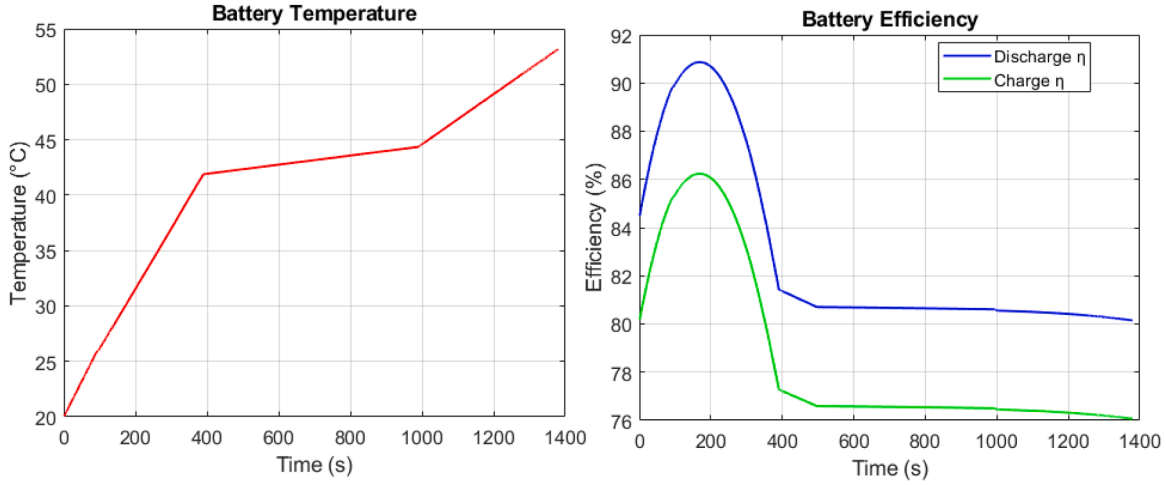


Fig. 13. Battery temperature rise and efficiency profiles (discharge and charge) under the proposed hierarchical MPC-guided SAC strategy.

characterized by higher SFC and lower efficiency, in order to satisfy overall power balance and state of charge requirements and constraints. As expected, the negative dependence between SFC and efficiency during the mission period verifies a positive dependence between control actions, fuel flow dynamics, and turbine thermodynamic properties.

4.3. Performance comparison with conventional MPC

This section compares the proposed hierarchical MPC-SAC framework with an MPC-only baseline, focusing on energy and environmental performance. The analysis emphasizes fuel consumption, CO_2 emissions

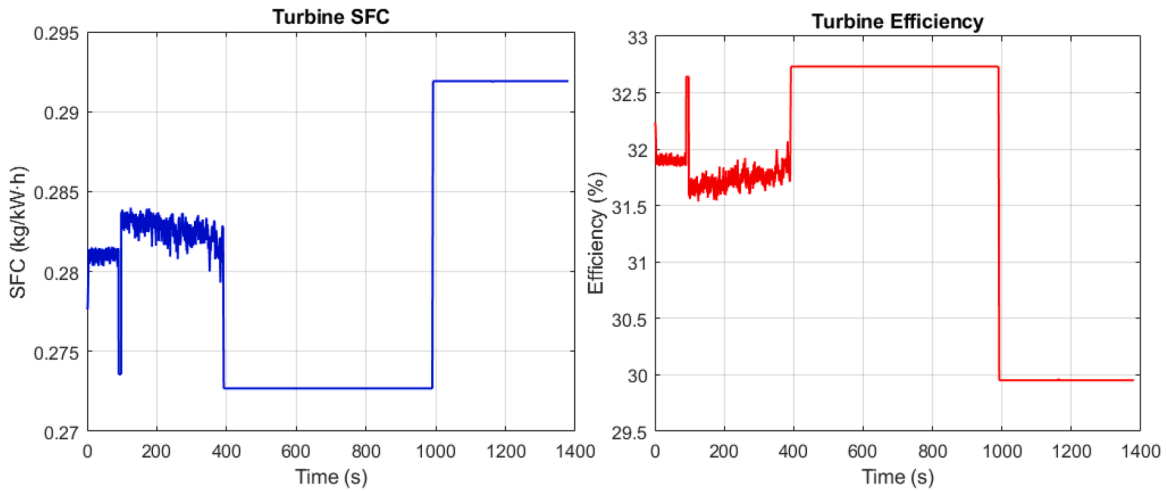


Fig. 14. Turbine efficiency and fuel consumption response under hybrid MPC-SAC energy management.

for both Jet-A and SAF, and adherence to battery fraction target levels (%), assessing whether the hybrid strategy delivers improved efficiency and sustainability while maintaining equivalent mission performance. It's worth mentioning that the performance comparison in Section 4.3 focuses primarily on fuel consumption and emissions between the proposed hybrid MPC-SAC approach and a conventional MPC baseline. This choice is deliberate. Conventional MPC provides explicit constraint handling, recursive feasibility guarantees (under appropriate conditions), and high interpretability, which are essential in the targeted safety-critical or real-time control setting. In contrast, pure model-free reinforcement learning methods such as SAC typically lack built-in hard constraint enforcement and may exhibit constraint violations or unstable behavior, especially during exploration and deployment in high-stakes applications. By comparing directly against MPC baseline, we can clearly isolate and demonstrate the incremental benefits introduced by the proposed MPC-guided learning mechanism while retaining the desirable safety and interpretability properties of model-based control. Although a full comparison of power fractions including the standalone SAC agent has already been presented earlier, we include a brief cross-reference and trade-off discussion here for completeness with respect to learning-based methods mentioned in the Introduction.

Table 6 presents a detailed comparison of fuel consumption and CO_2 emissions between the baseline MPC-only strategy and the proposed Hybrid MPC-SAC approach for a hybrid-electric distributed propulsion aircraft under three power source configurations: Pure-electric, conventional Jet-A fuel, and sustainable aviation fuel (SAF). All strategies and configurations complete the mission in an identical flight time of 0.38 h, ensuring fair performance benchmarking. For Jet-A fuel, the hybrid MPC-SAC reduces fuel usage from 10.60 kg (MPC-only) to 9.99 kg, achieving a 5.75% improvement, while CO_2 emissions drop from 33.38 kg to 31.47 kg (-5.72%). Similarly, with SAF, the hybrid strategy lowers fuel consumption from 10.60 kg to 9.99 kg (-5.75%) and CO_2 emissions from 10.01 kg to 9.44 kg (-5.69%). Notably, SAF delivers approximately 70% lower CO_2 emissions than Jet-A across both control strategies (MPC-only: 10.01 kg against 33.38 kg; Hybrid: 9.44 kg against 31.47 kg), underscoring the synergistic benefits of advanced control and sustainable fuels for reducing environmental impact without compromising mission duration.

This is evident in Fig. 15 which shows cumulative fuel consumption (kg) for MPC-only and hybrid MPC-SAC strategies. MPC-only is depicted by a red line, while hybrid MPC-SAC is shown as a blue line. Both trajectories start at zero and exhibit a near-linear upward trend with slight concavity, reflecting a gradually increasing fuel consumption rate. MPC-SAC consistently remains below MPC-only, indicating systematic fuel savings from improved policy optimality and stochastic handling. Early in the simulation, the curves overlap, but separation grows in the mid-to-late phases (400–1400 s), yielding cumulative benefits. By 1368 s, both reach approximately 9.99 and 10.6 kg, with hybrid MPC-SAC achieving a modest 2-5% reduction, demonstrating incremental efficiency gains without altering the overall consumption trend.

To determine the environmental impact and sustainability of the aircraft's energy management system, the cumulative CO_2 emission profiles are assessed and reveal that the MPC-SAC architecture offers a tangible advantage over the traditional MPC-only baseline, as shown in Fig. 16. While the adoption of Sustainable Aviation Fuel (SAF) provides the most substantial reduction in total carbon output, the integration of the Soft Actor-Critic (SAC) agent further optimizes these gains. Visually, the MPC-SAC trajectories for both Jet-A and SAF consistently track below their MPC-only counterparts, demonstrating that the reinforcement learning agent has successfully learned an energy-sharing policy that minimizes fuel consumption. By the end of the 1368 s mission, the hybrid controller achieves a lower total emission state without violating the aircraft's operational constraints. This indicates that while the MPC ensures immediate physical safety, the SAC agent adds a layer of long-term environmental foresight, prioritizing power-split configurations that yield superior fuel economy. Ultimately, these results suggest that the MPC-SAC framework is the superior implementation for missions where environmental impact and fuel conservation are prioritized alongside mission-critical performance. Recent hybrid MPC-RL studies reinforce this positioning. Ahmed et al. [41] showed that a

Table 6
Comparison of control strategy for hybrid-electric propulsion fuel consumption and CO_2 emissions.

Configuration		MPC only		Hybrid MPC-SAC		Δ%		Units
	Pure-Electric	Jet-A	SAF	Jet-A	SAF	Jet-A	SAF	
Flight time	0.38	0.38	0.38	0.38	0.38			hr
Fuel used	-	10.60	10.60	9.99	9.99	-5.75	-5.75	kg
CO_2 emissions	-	33.38	10.01	31.47	9.44	-5.72	-5.69	kg

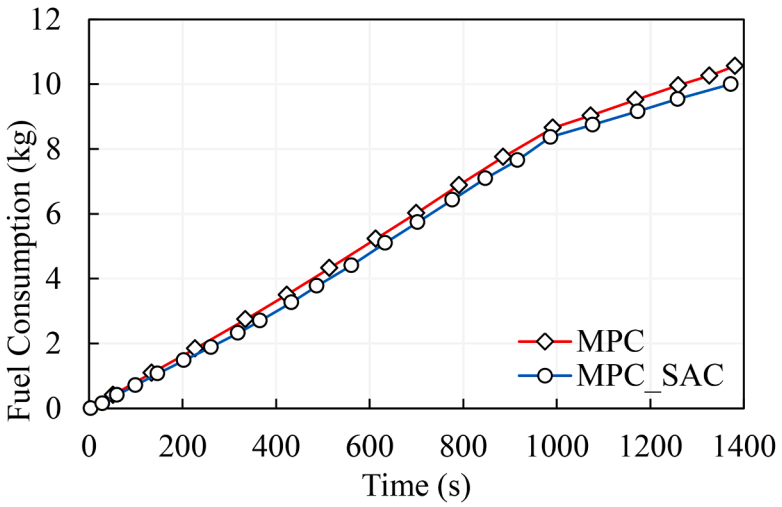


Fig. 15. Cumulative fuel consumption comparison between conventional MPC and hybrid MPC-SAC.

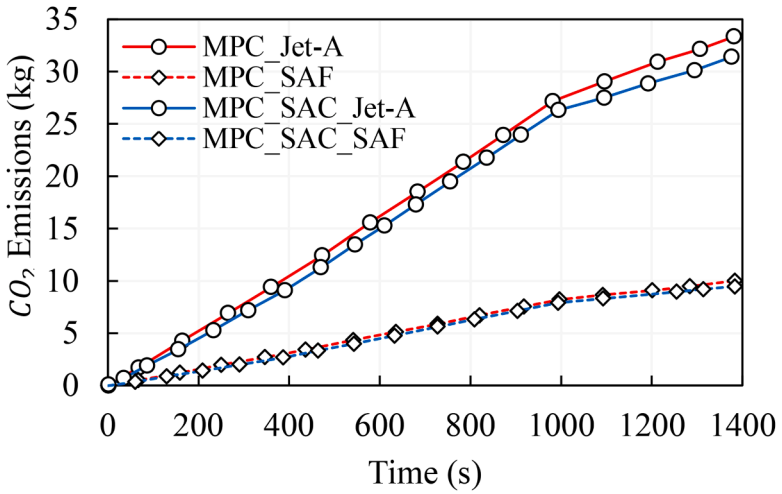


Fig. 16. Cumulative CO_2 emissions comparison across Jet-A and SAF Fuel types under MPC and Hybrid MPC-SAC control frameworks.

DeepONet-guided TD3 hybrid significantly outperforms pure TD3 on chaotic systems, achieving about 80% lower variance, more homogeneous behavior, and superior constraint satisfaction by blending offline model-based generalization with online RL adaptation. Similarly, Kostelac et al. [42] demonstrated on a nonlinear aeroelastic wing that their MPC-guided RL with Lipschitz safety filter yields the best trade-off: fewer state excursions, reduced actuator effort, and stronger disturbance rejection compared to pure RL (which reacts aggressively with high wear) and standalone MPC. These works highlight the consistent advantages of hybrid approaches over pure learning-based methods in safety-critical control.

In evaluating the hierarchical energy management system, the comparative results between the MPC-only and MPC-SAC battery fraction implementations in Fig. 17 reveal a clear performance advantage for the hybrid reinforcement learning approach. Visually, the MPC-only model predictions demonstrate near-perfect alignment with the reference trajectories, ensuring that thrust and power commands are met with minimal error. Conversely, the MPC-SAC trends exhibit a systematic divergence from the targets. This indicates that while the SAC agent is guided by the MPC safety layer, its internal reward structure encourages the controller to deviate from the nominal path to prioritize overall system health or secondary efficiency objectives. Specifically, the SAC agent has learned that following the exact target is less efficient than following a modified, learned trajectory that prioritizes long-term goals, such as thermal management or battery longevity, over immediate setpoint tracking. While MPC-only is superior for mission precision, the MPC-SAC results illustrate the agent's tendency to sacrifice short-term tracking accuracy for a more flexible, learned operational strategy that enhances the aircraft's total mission endurance. This trade-off suggests that the SAC-enhanced controller is more "intelligent" in its resource allocation, even if it appears less disciplined in following the target commands.

5. Conclusions

In conclusion, this study establishes that the hierarchical model predictive control-guided reinforcement learning framework provides an effective approach to energy management in hybrid-electric distributed propulsion aircraft. This integration yields balanced designs with reduced emissions, enhanced fuel efficiency, extended battery longevity, improved flight performance, and robust real-time constraint satisfaction across diverse operating conditions. Below are the key conclusions from the studies:

- The hierarchical MPC-guided SAC framework demonstrates effective and stable control, with bounded SAC actions, adaptive integration of MPC and RL, sustained hierarchical guidance, and strong

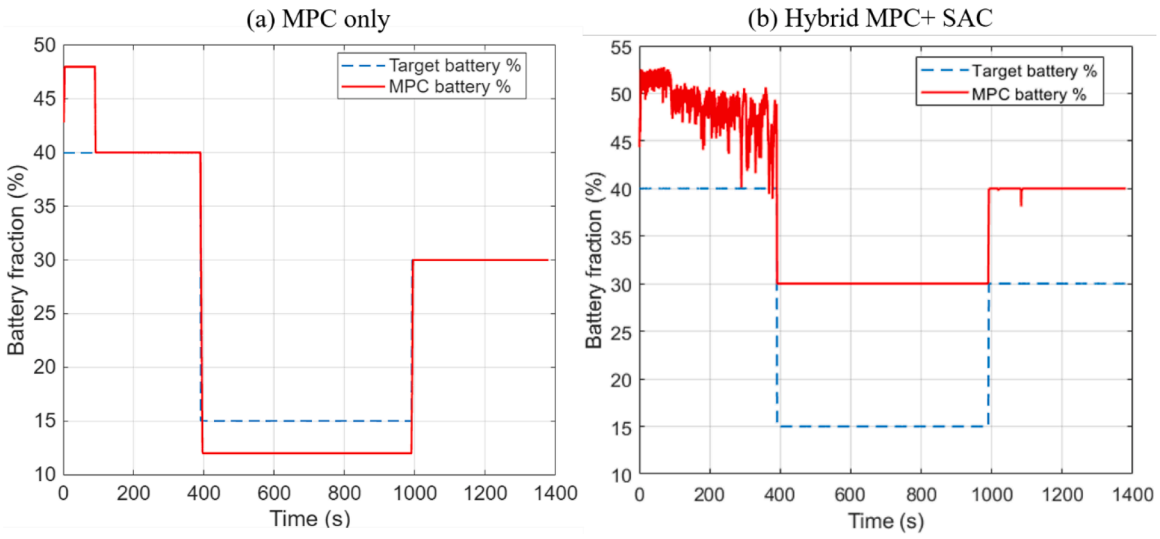


Fig. 17. Performance comparison of MPC-only and MPC-SAC architectures for hybrid-electric DEP Aircraft Energy Management and Fuel Optimization.

performance across stability (0.97), tracking (0.90), efficiency (0.80), and constraint satisfaction (0.84), achieving an overall score of 0.88.

- The results show that SAC corrections remain predominantly small, peaking around 0.3-0.4 in magnitude, with near-zero deviations during take-off, moderate spread in climb and descent, and dense concentration during cruise. Frequent re-planning triggered by >10% deviations indicates active responsiveness, while control authority transitions from MPC-dominant early to balanced MPC-SAC weighting mid-mission, supporting robust and adaptive hierarchical control.
- The results show that in-flight battery recharging occurs during cruise and descent along with asymmetric regeneration. SOC decreases steadily during take-off and climb, stabilizes during cruise, and recovers progressively during descent, with a single extended charging event delivering about 9.79 kWh at a peak power of about 103 kW. The hierarchical controller leverages these low-demand phases to maximize SOC recovery without compromising thrust, demonstrating effective energy recuperation, improved mission efficiency, and reduced fuel consumption.
- With regards to efficient energy management, the hierarchical MPC-guided SAC strategy achieves a 5.75% reduction in fuel consumption and approximately 5.7% lower CO_2 emissions compared to MPC-only, for both Jet-A and SAF, while maintaining identical 0.38-hour flight time. SAF yields substantially lower CO_2 emissions than Jet-A in both strategies.

While the results demonstrate the superior performance of the hybrid hierarchical MPC-SAC framework, the approach is not without its limitations. Further works will consider real-time implementation of hierarchical MPC-RL, which requires computationally efficient solvers for onboard deployment while learning from real flight data. This enables policies to bridge the simulation-to-reality gap. Furthermore, enhancing robustness to uncertainties and faults ensures safe, adaptive energy management under variable conditions, system degradation, and unexpected disturbances in hybrid-electric distributed propulsion aircraft.

CRediT authorship contribution statement

Richard Amankwa Adjei: Writing – review & editing, Writing – original draft, Supervision, Software, Formal analysis, Data curation, Conceptualization. **Da Xie:** Writing – review & editing, Software, Methodology, Formal analysis. **Shiao Li:** Writing – review & editing, Software, Methodology, Formal analysis. **Yong Ren:** Writing – review & editing, Supervision, Resources, Methodology. **Christos Spitas:** Writing – review & editing, Supervision, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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